Examples of Perverse Coxeter and Perverse Schobers

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Contents

1 Coxeter complexes

1.1 Coxeter complex and Tits product

Let (W,S) be a Coxeter system, namely S is a finite set and W is a group generated by elements $\{s_i\}_{i\in S}$ indexed by S and subject to the following relations

$$(s_i s_j)^{m_{ij}} = \mathrm{id}, \quad \forall i,j \in S, m_{ij} = m_{ji} \in \{2,3,4,\dots\} \cup \{\infty\}\,; m_{ii} = 1$$

Assume $V = V_{\mathbb{R}}$ is a Euclidean vector space, an *affine reflection* on V is of the form $v \mapsto v - 2\langle v, n\rangle n + 2\gamma n$, where n is a normalized vector perpendicular to some hyperplane in V that crosses the origin and $\gamma \in \mathbb{R}$. One can consider *affine reflection group* W that is generated by finitely many affine reflections on V such that $\{x \in W \mid F_1 \cap w \cdot F_2\}$ has finite cardinality for any choice of compact subsets $F_1, F_2 \subseteq V$. It can be shown that W is a Coxeter group and associated with such W, one can consider the hyperplane arrangement on V so that each element $w \in W$ determines a reflecting hyperplane H_w. The number m_{ij} is determined by angle between reflecting hyperplanes H_i and H_j.
Conversely, starting with a Coxeter system, one can associate it with a hyperplane arrangement so that its affine reflection group is the Coxeter group. Moreover, those hyperplanes cut V into a union of facets and realize V as a simplicial complex called *Coxeter complex*, denoted by $\mathrm{Cox}(W,S)$ The Coxeter group W acts on the Coxeter complex via reflection, for each subset $I \subseteq S$, let W_I be the subgroup of W generated by I with the same relation, it is called the parabolic subgroup of W. Moreover, the Coxeter complex can be realized as

$$\mathrm{Cox}(W,S) = \bigcup_{I\subseteq S} W/W_I$$

where the facets are labelled by cosets wW_I and inclusion of facets are inverse inclusion of cosets

$$wW_J \leqslant vW_K \iff wW_J \supseteq vW_K$$

Let $\mathcal{H}$ be a hyperplane arrangement on V which is locally finite, i.e. $\forall v \in V, \sharp\,(B_\epsilon(v) \cap \mathcal{H}) < \infty$.

Definition 1. We define an operation on $\Sigma(\mathcal{H})$: for $F, G \in \Sigma(\mathcal{H})$, choose any two points $\lambda \in F, \lambda' \in G$. Consider the segment in V connects F and G given by $(1-t)\lambda + t\lambda', 0 \leqslant t \leqslant 1$, define the **Tits product** $F \circledcirc G$ to be the first facet the segment meets when $0 < t < 1$.

Proposition 2. *The above definition does not depend on choice of λ, λ'. The Tits product $\circledcirc$ defines a semigroup structure on $\Sigma(\mathcal{H})$, in particular, if all hyperplanes in $\mathcal{H}$ intersects at one point, then $(\Sigma(\mathcal{H}), \circledcirc)$ is a monoid.*

In [KSSY24], we considered the case that $(W, \mathcal{I}_G)$ is the Coxeter system with respect to a complex reductive group G (assumed to be semisimple for simplicity) with fixed Borel B and a maximal torus $T \subseteq B$, therefore W will be the Weyl group of G and $\mathcal{I}_G$ is the vertices of the Dynkin diagram of G.
Denote by $\Phi, \check{\Phi}$ the root (resp. coroot) system of G and $\mathfrak{h}_{\mathbb{R}}$ the real form of $\mathfrak{h} = \mathrm{Lie}(T)$ such that $\mathrm{Span}(\check{\Phi}) \subseteq \mathfrak{h}_{\mathbb{R}}$. The corresponding hyperplane arrangement will be root hyperplane arrangement on $\mathfrak{h}_{\mathbb{R}}$. The facets of the root

hyperplane arrangement will be in correspondence with

$$\mathcal{P}_{G,T} = \{\text{parabolic subalgebras of } \mathfrak{g} = \mathrm{Lie}(G) \text{ containing } \mathfrak{h}\}.$$

For $\mathfrak{p} \in \mathcal{P}_{G,T}$, denote by $C_{\mathfrak{p}}^{\circ}$ the corresponding facet. The closure of $C_{\mathfrak{p}}^{\circ}$ would be $C_{\mathfrak{p}} := \overline{C_{\mathfrak{p}}^{\circ}} = \bigcup_{\mathfrak{q} \supseteq \mathfrak{p}} C_{\mathfrak{q}}^{\circ}$. Each facet can be endowed with a *color*, given by $I \subseteq \mathcal{I}_G$. More explicitly, there is a bijection:

$$\{I \subseteq \mathcal{I}_G\} \overset{\text{bij}}{\leftrightarrow} \mathcal{P}_{G,B} = \{\text{parabolic subalgebras of } \mathfrak{g} = \mathrm{Lie}(G) \text{ containing } \mathfrak{b} = \mathrm{Lie}(B)\}$$

The parabolic $\mathfrak{p}_I \in \mathcal{P}_{G,B}$ is called the standard parabolic and $\mathfrak{p} \in \mathcal{P}_{G,T}$ is said to have color I if $\mathfrak{p}$ is conjugate to $\mathfrak{p}_I$ by some element $w \in W$, or equivalently $C_{\mathfrak{p}}^{\circ} = w \cdot C_{\mathfrak{p}_I}^{\circ}$. Let $\mathcal{L}_{G,T}$ be the set of levi subalgebras of $\mathfrak{g}$ that contain $\mathfrak{h}$, then the flats in $\mathrm{Cox}(W, \mathcal{I}_G)$ are in bijection with $\mathcal{L}_{G,T}$, in particular the map $\mathcal{P}_{G,T} \to \mathcal{L}_{G,T} : \mathfrak{p} \mapsto \mathfrak{l}_{\mathfrak{p}} := \mathrm{Levi}(\mathfrak{p})$ corresponds to the operation $C_{\mathfrak{p}}^{\circ} \mapsto \mathrm{Span}_{\mathbb{R}}\left(C_{\mathfrak{p}}^{\circ}\right)$.
Also the Tits product in this case has Lie theoretic meaning, let $\mathfrak{p}, \mathfrak{q} \in \mathcal{P}_{G,T}$ with $C_{\mathfrak{p}}^{\circ} \circledcirc C_{\mathfrak{q}}^{\circ} = C_{\mathfrak{p} \circledcirc \mathfrak{q}}^{\circ}$, where the notion $\mathfrak{p} \circledcirc \mathfrak{q}$ is the element in $\mathcal{P}_{G,T}$ such the formula holds.

Lemma 3. *We have $\mathfrak{p} \circledcirc \mathfrak{q} \cong \mathfrak{p} \cap \mathfrak{q} + \mathfrak{n}_{\mathfrak{p}} \in \mathcal{P}_{G,T}$, where $\mathfrak{n}_{\mathfrak{p}}$ is the nilpotent radical of $\mathfrak{p}$.*

One can observe that $\mathfrak{p} \circledcirc \mathfrak{q} \neq \mathfrak{q} \circledcirc \mathfrak{p}$ in general, however their underlying Levi subalgebras are the same, in other word, the facets $C_{\mathfrak{p}}^{\circ} \circledcirc C_{\mathfrak{q}}^{\circ}$ and $C_{\mathfrak{q}}^{\circ} \circledcirc C_{\mathfrak{p}}^{\circ}$ are in the same flat.

Example 4. Let us consider the Coxeter complex of type A_2, which is depicted as following:

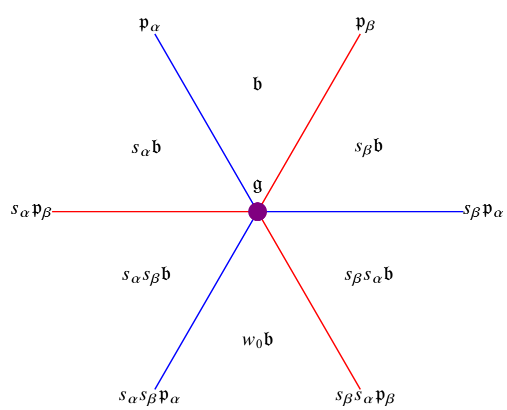


Figure 1: Coxeter complex of type A_2

There are 6 dimension = 2 facets of color $\emptyset$, 6 dimension = 1 facets, 3 of which are of color α, the other 3 are of color β and 1 dimension = 0 facet, corresponds to $\mathfrak{g}$, with color $\mathcal{I}_{A_2}$.

1.2 Braid groupoid and perverse Coxeter category

Given a stratification of topological space X by poset P, it is known that if the stratification (X, P) is conic, then the category of constructible sheaves on X with respect to this stratification is determined by the exit category or the entry category of (X, P).

Theorem 5. *[L17, Theorem A.9.3] There is an equivalence of categories*

$$\mathrm{Const}_P(X) \cong \mathrm{Func}(\mathrm{Exit}(X,P), \mathbf{Spc})$$

In general it is more difficult to count-in both exit and entry paths. When the stratified space (X, P) is given by hyperplane arrangement, Kapranov and Schechtman give a description of perverse sheaves on (X, P) by considering both exit and entry paths.

Theorem 6. *[KS16] There is an equivalence of categories*

$$\mathrm{Perv}_P(X, \mathcal{A}) \cong \mathrm{Func}\left(\mathrm{Quiv}^{(2)}(X, P), \mathcal{A}\right)$$

between perverse sheaves valued in abelian category $\mathcal{A}$ and functors from $\mathrm{Quiv}^{(2)}$ *to $\mathcal{A}$.*

One example of hyperplane arrangement is given by root hyperplanes on $\mathfrak{h}_{\mathbb{R}}$. In [KSSY24], one can consider the **double incidence category** Q with objects being facets in the Coxeter complex, i.e. $\mathrm{Ob}(Q) = \mathcal{P}_{G,T}$. Recall the poset structure on $\mathcal{P}_{G,T}$ is given by the inclusion relation between parabolic subalgebras, so we write $\mathrm{Res}^{\mathfrak{q}}_{\mathfrak{p}}$ (resp. $\mathrm{Ind}^{\mathfrak{p}}_{\mathfrak{q}}$) for the exit path (resp. entry path) for any pair $(\mathfrak{p}, \mathfrak{q})$ such that $C^{\circ}_{\mathfrak{p}} \subseteq C_{\mathfrak{q}}$. Accordingly, the morphisms in Q are generated by $\mathrm{Res}^{\mathfrak{q}}_{\mathfrak{p}}$ and $\mathrm{Ind}^{\mathfrak{p}}_{\mathfrak{q}}$ subject to relations:

- $\mathrm{Ind}^{\mathfrak{p}}_{\mathfrak{p}} = \mathrm{Res}^{\mathfrak{p}}_{\mathfrak{p}} = \mathrm{Id}_{\mathfrak{p}}$
- For any $\mathfrak{p} \subseteq \mathfrak{p}' \subseteq \mathfrak{p}''$, $\mathrm{Ind}^{\mathfrak{p}''}_{\mathfrak{p}'} \circ \mathrm{Ind}^{\mathfrak{p}'}_{\mathfrak{p}} = \mathrm{Ind}^{\mathfrak{p}''}_{\mathfrak{p}}$; $\quad \mathrm{Res}^{\mathfrak{p}}_{\mathfrak{p}'} \circ \mathrm{Res}^{\mathfrak{p}'}_{\mathfrak{p}''} = \mathrm{Res}^{\mathfrak{p}}_{\mathfrak{p}''}$
- $\mathrm{Res}^{\mathfrak{q}}_{\mathfrak{p}} \circ \mathrm{Ind}^{\mathfrak{p}}_{\mathfrak{q}} = \mathrm{Id}_{\mathfrak{q}} \quad \forall \mathfrak{q} \subseteq \mathfrak{p}$

Under above three relations, one can define for any $\mathfrak{p}, \mathfrak{p}' \in \mathcal{P}_{G,T}$ a morphism $\tau^{\mathfrak{p}'}_{\mathfrak{p}} := \mathrm{Res}^{\mathfrak{p}'}_{\mathfrak{q}} \circ \mathrm{Ind}^{\mathfrak{q}}_{\mathfrak{p}}$ which does not depend on choice of $\mathfrak{q}$. The rest of relations in Q are

- If $\mathfrak{p}, \mathfrak{p}', \mathfrak{p}''$ are collinear, i.e. there are points $p \in C^{\circ}_{\mathfrak{p}}, p' \in C^{\circ}_{\mathfrak{p}'}, p'' \in C^{\circ}_{\mathfrak{p}''}$ so that p' is in the line segment $[p, p'']$, then $\tau^{\mathfrak{p}''}_{\mathfrak{p}} = \tau^{\mathfrak{p}''}_{\mathfrak{p}'} \circ \tau^{\mathfrak{p}'}_{\mathfrak{p}}$
- If $C_{\mathfrak{p}'} \supseteq C_{\mathfrak{p}} \subseteq C_{\mathfrak{p}''}$ as codimension 1 subspace and $C_{\mathfrak{p}'}, C_{\mathfrak{p}''}$ are opposite in the $\mathrm{Span}(C_{\mathfrak{p}'})$, then both $\tau^{\mathfrak{p}''}_{\mathfrak{p}'}$ and $\tau^{\mathfrak{p}'}_{\mathfrak{p}''}$ are invertible.

The operators $\tau^{\mathfrak{p}'}_{\mathfrak{p}}$ are called **braiding operators/wall-crossing operators**, which can be seen as the braiding of the fundamental groupoid associated with root hyperplane arrangement. The construction of fundamental groupoid for chambers has been studied by Deligne [D72], see also [KSSY24, Section 2.4] for an explicit description. In the left of Figure 2, the orange path represents a braiding $\tau^{s_\alpha s_\beta \mathfrak{b}}_{\mathfrak{b}}$, which can also be decomposed into the composition $\tau^{s_\alpha s_\beta \mathfrak{b}}_{s_\alpha \mathfrak{b}} \circ \tau^{s_\alpha \mathfrak{b}}_{\mathfrak{b}}$. In addition, in [KSSY24], we introduce the W-**orbit category** of Q (see [KSSY24, Section A.6] for a definition of orbit category), where the objects are W-orbits of $\mathcal{P}_{G,T}$, that corresponds to colors $I \subseteq \mathcal{I}_G$ and the Hom-space of two orbits I_1, I_2 are W-invariant families of $\mathrm{Hom}_Q(\mathfrak{p}_1, \mathfrak{p}_2)$, where $\mathfrak{p}_1$ (resp. $\mathfrak{p}_2$) runs over elements in the orbit I_1 (resp. I_2). We call the resulting orbit category the **perverse Coxeter category** C (P-Coxeter in short). The name is originated from the following theorem

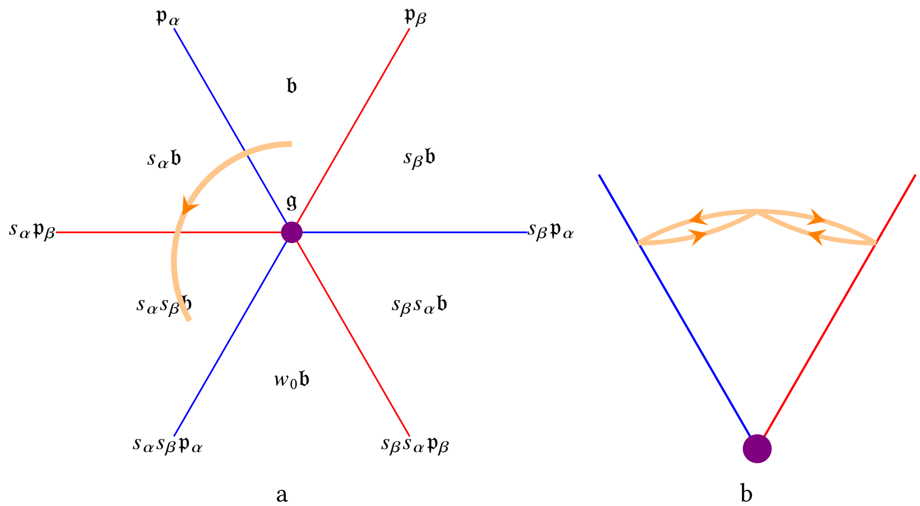


Figure 2: Braidings in Coxeter complex of type A_2

Theorem 7 ([KSSY24])**.** *There is an equivalence*

$$\mathrm{Perv}_{\mathcal{A}}(\mathfrak{h}//W, \mathcal{A}) = \mathrm{Func}(\mathcal{C}, \mathcal{A})$$

Heuristically one can think of the orbit space of the Coxeter complex by folding the chambers, as in the right figure. The braiding operators $\tau_{\mathfrak{p}}^{\mathfrak{q}}$ now become the monodromies in the orbit space. For example the operator $\tau_{\mathfrak{b}}^{s_\alpha s_\beta \mathfrak{b}}$ becomes the monodromy in the Figure 2(b).

2 Functions on commuting varieties and Langlands parameters

Let X_0 be a geometrically connected, smooth projective curve over $\mathbb{F}_q$ and $X := X_0 \times_{\mathbb{F}_q} \bar{\mathbb{F}}_q$ be the base-change to algebraically closed field. Let G be a split reductive group over $\mathbb{F}_q$ and its Langlands dual group $\check{G}$ over $\mathbb{C}$. We also fix an isomorphism $\overline{\mathbb{Q}}_\ell \cong \mathbb{C}$, where $\ell \neq \mathrm{char}(q)$ a prime number. In this section, we will compare several vector spaces and reveal an arithmetic/decategorified version of restricted geometric Langlands correspondence.

Let $\sigma \in \mathrm{LS}^{\mathrm{restr}}_{\check{G}}(X)$, then $H^{\bullet}_{et}(X, \check{\mathfrak{g}}_\sigma)$ is a dg-algebra. One can consider the following commuting variety:

$$\mathrm{Comm}_\sigma := \left\{ u \in H^1_{et}(X, \check{\mathfrak{g}}_\sigma) \mid [u,u] = 0 \in H^2_{et}(X, \check{\mathfrak{g}}_\sigma) \right\}$$

It is equipped with an action of $\mathrm{GSp}(X) \times \check{G}$, where $\mathrm{GSp}(X) := \mathrm{GSp}(H^1_{et}(X, \overline{\mathbb{Q}}_\ell))$-action is induced from the natural symplectic structure of $H^1_{et}(X, \overline{\mathbb{Q}}_\ell)$ (which is a $2g_X$-dimensional vector space) and the $\check{G}$-action is given by adjoint action on $\check{\mathfrak{g}}_\sigma$. One can consider the equivariant Grothendieck group $K^{\mathrm{GSp}\times\check{G}}(\mathrm{Comm}_\sigma)$.

Proposition 8. *Take σ to be the trivial local system so that $\check{\mathfrak{g}}_\sigma = \check{\mathfrak{g}}$, then there exists a morphism:*

$$K^{\mathrm{GSp}\times\check{L}}(\mathrm{Comm}_{\check{\mathfrak{l}}}) \longrightarrow K^{\mathrm{GSp}\times\check{G}}(\mathrm{Comm}_{\check{\mathfrak{g}}})$$

where $\mathrm{Comm}_{\check{\mathfrak{g}}}$ *denotes for* $\mathrm{Comm}_{\mathrm{triv}}$ *(in which case $\check{\mathfrak{g}}_\sigma \cong \check{\mathfrak{g}}$), so does* $\mathrm{Comm}_{\check{\mathfrak{l}}}$.

Proof: The proof follows from the construction in [SV12, section 2]. Let Y_0 be L-variety and $Y_1 := G \times_P Y_0$ the induced G-variety, where $P = LU$ is parabolic subgroup containing L. Assume there is a diagram of smooth varieties $Y_0 \xleftarrow{p} V \xrightarrow{q} Y$ such that p is a fibration of L-varieties and q is a closed embedding into G-variety. Then we can get a diagram of G-varieties:

$$Y_1 \xleftarrow{f} W = G \times_P V \xrightarrow{g} Y$$

where both maps f and g are equivariant and $W \xrightarrow{(f,g)} Y_1 \times Y$ is a closed embedding. Denote the conormal bundle of W relative to $Y_1 \times Y$ by $Z := T^*_W(Y_1 \times Y)$, it fits into the following diagram

$$\begin{tikzcd} T^*Y_1 & Z \arrow[l, "\phi"'] \arrow[r, "\psi"] & T^*Y \\ T^*_G Y_1 \arrow[u] & Z_G := Z \cap (T^*_G Y_1 \times T^*_G Y) \arrow[l, "\phi_G"'] \arrow[r, "\psi_G"] \arrow[u] & T^*_G Y \arrow[u] \end{tikzcd}$$

where $T^*_G Y_1$ (resp. $T^*_G Y$) denotes also the conormal bundle. One can show ([SV12, Lemma 2.3]) that both maps ϕ, ψ are regular and ψ is proper. Thus we can consider $R\psi_* : K^G(Z) \to K^G(T^*Y)$ and $L\phi^* : K^G(T^*Y_1) \to K^G(Z)$ which induces $R\psi_* : K^G(Z_G) \to K^G(T^*_G Y)$ and $L\phi^* : K^G(T^*_G Y_1) \to K^G(Z_G)$. Composing the two map we obtain

$$R\psi_* L\phi^* : K^G(T^*_G Y_1) \cong K^L(T^*_L Y_0) \longrightarrow K^G(T^*_G Y).$$

Now we can replace G above to be our $\check{G}$ and $P = LU$ to be $\check{P} = \check{L}\check{U}$. We take $Y_0 = \check{\mathfrak{l}}^{gx}, V = \check{\mathfrak{p}}^{gx}$ and $Y = \check{\mathfrak{g}}^{gx}$, then we get a parabolic induction diagram of Lie algebras $Y_0 \xleftarrow{p} V \xrightarrow{q} Y$, in particular, $T^*Y \cong \check{\mathfrak{g}}^{gx} \times \check{\mathfrak{g}}^{gx}$ and the conormal bundle $T^*_{\check{G}}Y \cong \mathrm{Comm}_{\check{\mathfrak{g}}}$. On the other hand we have (see [SV12, Lemma 2.4])

$$T^*Y_1 \cong \check{G} \times_{\check{P}} \left\{(u,v) \in H^1_{\mathrm{et}}(X, \check{\mathfrak{l}}) \oplus H^2_{\mathrm{et}}(X, \check{\mathfrak{p}}) \mid v|_{\check{\mathfrak{l}}} = [u,u]\right\}$$

which implies $T^*_{\check{G}}Y_1 = \check{G} \times_{\check{P}} \mathrm{Comm}_{\check{\mathfrak{l}}}$. Apply $R\psi_* L\phi^*$ we obtain the map we want. $\square$

QUESTION: How can one construct a morphism $K^{\mathrm{GSp}\times\check{G}}(\mathrm{Comm}_{\check{\mathfrak{g}}}) \to K^{\mathrm{GSp}\times\check{L}}(\mathrm{Comm}_{\check{\mathfrak{l}}})$, in a natural way?

On the other hand, we can consider the so-called **spherical Hall algebra** associated with X. We denote by AF_G (resp. $\mathrm{AF}_{G,c}$) the space of automorphic functions (resp. those with compact support) $\mathrm{Funct}(\mathrm{Bun}_G(\mathbb{F}_q), \mathbb{C})$ (resp. $\mathrm{Funct}_c(\mathrm{Bun}_G(\mathbb{F}_q), \mathbb{C})$) Let P be a standard parabolic subgroup of G that contains Levi subgroup M. We have the following diagram of correspondence:

$$\mathrm{Bun}_L \xleftarrow{\mathfrak{q}_P} \mathrm{Bun}_P \xrightarrow{\mathfrak{p}_P} \mathrm{Bun}_G$$

that induces two operators

$$\begin{aligned} \mathrm{Eis}^G_{L,P} &: \mathrm{AF}_L \to \mathrm{AF}_G : f \mapsto \mathfrak{p}_{P,!}\mathfrak{q}^*_P(f) \quad && \text{Eisenstein series} \\ \mathrm{CT}^L_{G,P} &: \mathrm{AF}_G \to \mathrm{AF}_L : f \mapsto \mathfrak{q}_{P,!}\mathfrak{p}^*_P(f) \quad && \text{constant term} \end{aligned}$$

Remark 9. The Eisenstein series $\mathrm{Eis}^G_{L,P}$ also sends $\mathrm{AF}_{L,c}$ to $\mathrm{AF}_{G,c}$, however the constant term $\mathrm{CT}^L_{G,P}$ does not preserve compact supportness. Assume we have inclusion of Levi subgroups $L \subseteq M$, with corresponding standard parabolic subgroups $Q \subseteq P$, then we can consider the following commutative diagram

$$\begin{tikzcd}
 & & \mathrm{Bun}_P \arrow[dl] \arrow[dr] \arrow[dll, bend right, "\mathfrak{q}_P"'] \arrow[drr, bend left, "\mathfrak{p}_P"] \arrow[dd, phantom, "\wedge", very near start] & & \\
 & \mathrm{Bun}_{M\cap P} \arrow[dl, "\mathfrak{q}_{M\cap P}"'] \arrow[dr, "\mathfrak{p}_{M\cap P}"] & & \mathrm{Bun}_Q \arrow[dl, "\mathfrak{q}_Q"'] \arrow[dr, "\mathfrak{p}_Q"] & \\
\mathrm{Bun}_L & & \mathrm{Bun}_M & & \mathrm{Bun}_G
\end{tikzcd}$$

We can deduce that $\mathrm{Eis}^G_{L,P} = \mathrm{Eis}^M_{L,M\cap P} \circ \mathrm{Eis}^G_{M,Q}; \mathrm{CT}^L_{G,P} = \mathrm{CT}^L_{M,M\cap P} \circ \mathrm{CT}^M_{G,Q}$

For torus $T \subseteq G$, we define the spherical part in $\mathrm{AF}_{T,c}$ to be locally constant functions, denoted by $\mathbf{U}_{X,H}$. Note that the connected components of Bun_T are given by coweights $\check{\Lambda}$, thus $\mathbf{U}_{X,T}$ is spanned by characteristic functions $\mathbf{1}_{\check{\lambda}}, \check{\lambda} \in \check{\Lambda}$. In particular we have $\mathbf{U}_{X,T} \cong \mathbb{C}\left[\check{\Lambda}\right]$. For Levi subgroup L, we define $\mathbf{U}_{X,L}$ to be the image of $\mathrm{Eis}_{T,L\cap B}$ over the spherical part $\mathbf{U}_{X,T}$. In particular, one can determine the formula of $\mathrm{Eis}^G_{L,P}$ and $\mathrm{CT}^L_{G,P}$ restricted to $\mathbf{U}_{X,L}$ and $\mathbf{U}_{X,G}$.

Proposition 10. *Let ζ_X be the Weil-zeta function associated with X, denote by $\widetilde{\zeta}_X(z) := \zeta_X(z)(1-qz)(1-qz^{-1})$,*

$e_X(z) := z^{1-g_X}\widetilde{\zeta}_X$ and $e_{\check{\lambda}} := e_X(\mathbf{1}_{\check{\lambda}})$. Define

$$\Psi_L^G : \mathbf{U}_L \to \mathbf{U}_G; f \mapsto \sum_{\sigma \in W_G/W_L} \sigma \cdot \left(\left(\prod_{\alpha \in \Phi_G^+ - \Phi_L^+} e_{\check{\alpha}} \right) \cdot f \right)$$

where $\Phi_L^+ \subseteq \Phi_G^+$ means the set of positive roots and $\check{\alpha}$ is the coweight corresponds to root α, thus the above operator depends on a choice of standard parabolics, we abbreviate as Ψ_L^G if there is no ambiguity. We have the following commutative diagram:

$$\begin{tikzcd}
\mathrm{AF}_{T,c} \arrow[r, "\mathrm{Eis}^L_{T,B\cap L}"] & \mathrm{AF}_{L,c} \arrow[r, "\mathrm{Eis}^G_{L,P}"] & \mathrm{AF}_{G,c} \\
\mathbf{U}_{X,T} \cong \mathbb{C}[\check{\Lambda}] \arrow[u, hook] \arrow[r, "\Psi^L_T"] \arrow[rr, bend right, "\Psi^G_T"'] & \mathbf{U}_{X,L} \arrow[u, hook] \arrow[r, "\Psi^G_L"] & \mathbf{U}_{X,G} \arrow[u, hook]
\end{tikzcd}$$

Proof: We will explain the formula in the context of restricted geometric Langlands. As is shown in [R], there is an isomorphism $\Gamma\left(\mathrm{LS}^{\mathrm{restr},\circ,\text{non-res}}_{\check{G}}, \omega_{\mathrm{LS}^{\mathrm{restr}}_{\check{G}}}\right) \cong \Gamma\left(\check{G} \overset{\mathrm{ad}}{/} \check{G}, \mathcal{O}_{\check{G} \overset{\mathrm{ad}}{/} \check{G}}\right)[\delta_{\check{G}}^{-1}]$, while the later is nothing but $\mathbb{C}[\check{\Lambda}]^{W_G}[\delta_{\check{G}}^{-1}]$. The Eisenstein seires on the level of groups, from $\check{T} \overset{\mathrm{ad}}{/} \check{T}$ to $\check{G} \overset{\mathrm{ad}}{/} \check{G}$ is given by Weyl character formula. The compatibility of Eisenstein series between $\mathrm{LS}^{\mathrm{restr},\circ}_{\check{G}}$ and $\check{G} \overset{\mathrm{ad}}{/} \check{G}$ is given by [R, Theorem 4.7.2.1]. Roughly speaking, we have the following diagram commutes

$$\begin{tikzcd}
\Gamma(\check{T} \overset{\mathrm{ad}}{/} \check{T}, \omega_{\check{T} \overset{\mathrm{ad}}{/} \check{T}}) \arrow[r, "\mathrm{Sym}"] & \Gamma(\check{G} \overset{\mathrm{ad}}{/} \check{G}, \omega_{\check{G} \overset{\mathrm{ad}}{/} \check{G}}) \arrow[d] \\
\Gamma\left(\mathrm{LS}^{\mathrm{restr},\circ}_{\check{T}}, \omega_{\mathrm{LS}^{\mathrm{restr},\circ}_{\check{T}}}\right) \arrow[u, "\cdot\left(\prod_{\alpha\in\Phi^+_G} e_{\check{\alpha}}\right)"] \arrow[r, "\Psi^G_T"] & \Gamma\left(\mathrm{LS}^{\mathrm{restr},\circ}_{\check{G}}, \omega_{\mathrm{LS}^{\mathrm{restr},\circ}_{\check{G}}}\right)
\end{tikzcd}$$

The formulas of Ψ_T^L and Ψ_T^G thus follow from the assumption of restricted geometric Langlands and its compatibility with Eisenstein series. Finally, one need to check that Ψ_L^G makes the diagram commute. It remains to show that for any $f \in \mathbf{U}_T$, we have the following:

$$\sum_{\sigma \in W_G/W_L} \sigma \cdot \left(\left(\prod_{\alpha \in \Phi_G^+ - \Phi_L^+} e_{\check{\alpha}} \right) \cdot \sum_{\sigma' \in W_L} \sigma' \cdot \left(\left(\prod_{\beta \in \Phi_L^+} e_{\check{\beta}} \right) \cdot f \right) \right) = \sum_{w \in W_G} w \cdot \left(\left(\prod_{\gamma \in \Phi_G^+} e_{\check{\gamma}} \right) \cdot f \right)$$

But it follows from simple combinatorics. $\square$

Proposition 11. *Let $\widehat{\mathbf{U}}_{X,L} \subseteq \mathrm{AF}_L$ be the formal completion of $\mathbf{U}_{X,L}$, define*

$$\Xi_G^L : \mathbf{U}_{X,G} \to \widehat{\mathbf{U}}_{X,L}; f \mapsto \left(\prod_{\alpha' \in \Phi_G^+ - \Phi_L^+} e_{\check{\alpha}'}^{-1} \right) \cdot f$$

where $e_{\check{\alpha}}^{-1}$ means $e_X(\mathbf{1}_{\check{\alpha}})^{-1}$. We have the following commutative diagram:

$$\begin{tikzcd}
\mathrm{AF}_{G,c} \arrow[r, "\mathrm{CT}_{G,Q}^M"] & \mathrm{AF}_M \arrow[r, "\mathrm{CT}_{M,M\cap P}^L"] & \mathrm{AF}_G \\
\mathbf{U}_{X,G} \arrow[u, hook] \arrow[r, "\Xi_G^M"] \arrow[rr, bend right, "\Xi_G^L"'] & \widehat{\mathbf{U}}_{X,M} \arrow[u, hook] \arrow[r, "\Xi_M^L"] & \widehat{\mathbf{U}}_{X,L} \arrow[u, hook]
\end{tikzcd}$$

Now let us the expression of composition of two operators $\Xi_G^M \circ \Psi_L^G$, we have the following diagram:

$$\begin{tikzcd}
\mathrm{AF}_{L,c} \arrow[r, "\mathrm{Eis}_{L,P}^G"] & \mathrm{AF}_{G,c} \arrow[r, "\mathrm{CT}_{G,Q}^M"] & \mathrm{AF}_M \\
\mathbf{U}_{X,L} \arrow[u, hook] \arrow[r, "\Psi_L^G"] & \mathbf{U}_{X,G} \arrow[u, hook] \arrow[r, "\Xi_G^M"] & \widehat{\mathbf{U}}_{X,M} \arrow[u, hook]
\end{tikzcd}$$

The composition in the top horizontal row also equals to the expression

$$\sum_{w \in W_L \backslash W_G / W_M} \mathrm{Eis}_{{}^wM}^M \circ R_w \circ \mathrm{CT}_L^{L^w}$$

which is called the Langlands formula. Similar thing happens on the lower horizontal row:

Proposition 12. *The composition $\Xi_G^M \circ \Psi_L^G$ equals to the sum*

$$\Xi_G^M \circ \Psi_L^G = \sum_{w \in W_L \backslash W_G / W_M} \Psi_{{}^wM}^M \circ \mathrm{Br}_w \circ \Xi_L^{L^w}$$

where we can write down explicitly the operator $\mathrm{Br}_w : f \mapsto \prod_{\beta \in w\Phi_G^- \cap \Phi_G^+} \frac{e_{-\check{\beta}}}{e_{\check{\beta}}} \cdot w(f)$, and we call the above formula the ***generalized Gindikin-Karpelevich formula****.*

3 Variations of Coxeter categories

3.1 Iwahori-Hecke algebra

Let (W, S) be a finitary Coxeter system. Recall that one can associate it with the Iwahori-Hecke algebra $\mathcal{H}_{(W,S)}$ with generators $T_s, s \in S$ and satisfy the following relations

$$\underbrace{T_sT_tT_s}_{m_{s,t}} = \underbrace{T_tT_sT_t}_{m_{s,t}} \quad \text{braid relations}$$
$$(T_s - q)(T_s + 1) = 0 \quad \text{quadratic relations}$$

Here q is a formal parameter. Specifying q at different value gives several variations of Hecke algebra. If we take $q \mapsto 1$, then the quadratic relation becomes $T_s^2 = 1$ and thus $\mathcal{H}^{q=1}_{(W,S)}$ can be identified with the group algebra of the Coxeter group W. In this case, we can consider the category $\mathcal{W}$ whose objects are finite subset $I \subseteq S$ and morphisms are identified with the double coset $\mathrm{Hom}_{\mathcal{W}}(I, J) = W_I\backslash W/W_J$. For two morphisms $\sigma_1 \in W_I\backslash W/W_J$ and $\sigma_2 \in W_J\backslash W/W_K$, their composition in $\mathcal{W}$ is defined to be $\sigma_1\sigma_2 \in W_I\backslash W/W_K$.

Lemma 13. *There is a canonical bijection:*

$$W_J\backslash W/W_K \overset{\text{bij}}{\simeq} \{w \in W \mid \ell(v'w) = \ell(v') + \ell(w), \ell(wv) = \ell(w) + \ell(v), \forall v' \in W_J, v \in W_K\}$$

The elements in the right-hand side are shortest representative of the double coset. Moreover, for any $\sigma \in W_J\backslash W/W_K$, there exists a unique ${}^{\sigma}J \subset J$ such that $W_{{}^{\sigma}J} = \sigma \cdot W_K \cap W_J$ and K^{σ} such that $W_{K^{\sigma}} = \sigma^{-1} \cdot W_J \cap W_K = \sigma^{-1} \cdot W_{{}^{\sigma}J}$.

For (W, S) Coxeter system associated with (G, T), with Δ_I the corresponding roots of $I \subseteq S$. Let $\sigma \in W_J\backslash W/W_K$ as above, then the Lemma 13 implies particularly that $\Delta_{{}^{\sigma}J} = \sigma \cdot \Delta_K \cap \Delta_J$ and $\Delta_{K^{\sigma}} = \sigma^{-1} \cdot \Delta_J \cap \Delta_K = \sigma^{-1} \cdot \Delta_{{}^{\sigma}J}$.

Lemma 14. *In the case where (W, S) is the Coxeter system associated with (G, T), assume that $\sigma \in W_J\backslash W/W_K$, then we have the Tits product $\mathfrak{p}_J \circledcirc \sigma \cdot \mathfrak{p}_K = \mathfrak{p}_{{}^{\sigma}J}$.*

Thus for any $\sigma \in W_J\backslash W/W_K$, we can associate it with a family $\tilde{\mathfrak{P}}_\sigma$ of morphisms in $\mathcal{Q}$:

$$\begin{aligned}\tilde{\mathfrak{P}}_\sigma(\mathfrak{p}_J) &:= \mathrm{Ind}^{\sigma\cdot\mathfrak{p}_K}_{\sigma\cdot\mathfrak{p}_K\circledcirc\mathfrak{p}_J} \circ \tau^{\sigma\cdot\mathfrak{p}_K\circledcirc\mathfrak{p}_J}_{\mathfrak{p}_J\circledcirc\sigma\cdot\mathfrak{p}_K} \circ \mathrm{Res}^{\mathfrak{p}_J\circledcirc\sigma\cdot\mathfrak{p}_K}_{\mathfrak{p}_J} \in \mathrm{Hom}_{\mathcal{Q}}(\mathfrak{p}_J, \sigma\cdot\mathfrak{p}_K),\\ \tilde{\mathfrak{P}}_\sigma(w\cdot\mathfrak{p}_J) &:= w(\mathfrak{P}_\sigma(\mathfrak{p}_J)) \in \mathrm{Hom}_{\mathcal{Q}}(w\cdot\mathfrak{p}_J, w\sigma\cdot\mathfrak{p}_K) \quad \forall w \in W\end{aligned}$$

Proposition 15. *For $\sigma_1 \in W_I\backslash W/W_J, \sigma_2 \in W_J\backslash W/W_K$, we have*

$$\tilde{\mathfrak{P}}_{\sigma_1\sigma_2} = \sigma_1(\tilde{\mathfrak{P}}_{\sigma_2}) \circ \tilde{\mathfrak{P}}_{\sigma_1}$$

Here we regard $\sigma_1\sigma_2$ as a double coset in $W_I\backslash W/W_K$.

Proof: We only need to verify the equality for $\mathfrak{p}_I$. The left hand side apply to $\mathfrak{p}_I$ is:

$$\mathrm{Ind}^{\sigma_1\sigma_2\cdot\mathfrak{p}_K}_{\sigma_1\sigma_2\cdot\mathfrak{p}_K\circledcirc\mathfrak{p}_I} \circ \tau^{\sigma_1\sigma_2\cdot\mathfrak{p}_K\circledcirc\mathfrak{p}_I}_{\mathfrak{p}_I\circledcirc\sigma_1\sigma_2\cdot\mathfrak{p}_K} \circ \mathrm{Res}^{\mathfrak{p}_I\circledcirc\sigma_1\sigma_2\cdot\mathfrak{p}_K}_{\mathfrak{p}_I}$$

while the right hand side is

$$\mathrm{Ind}^{\sigma_1\sigma_2\cdot\mathfrak{p}_K}_{\sigma_1\sigma_2\cdot\mathfrak{p}_K\circledcirc\sigma_1\mathfrak{p}_J}\circ\tau^{\sigma_1\sigma_2\cdot\mathfrak{p}_K\circledcirc\sigma_1\mathfrak{p}_J}_{\sigma_1\mathfrak{p}_J\circledcirc\sigma_1\sigma_2\cdot\mathfrak{p}_K}\circ\mathrm{Res}^{\sigma_1\mathfrak{p}_J\circledcirc\sigma_1\sigma_2\cdot\mathfrak{p}_K}_{\sigma_1\mathfrak{p}_J}\circ\mathrm{Ind}^{\sigma_1\cdot\mathfrak{p}_J}_{\sigma_1\cdot\mathfrak{p}_J\circledcirc\mathfrak{p}_I}\circ\tau^{\sigma_1\cdot\mathfrak{p}_J\circledcirc\mathfrak{p}_I}_{\mathfrak{p}_I\circledcirc\sigma_1\cdot\mathfrak{p}_J}\circ\mathrm{Res}^{\mathfrak{p}_I\circledcirc\sigma_1\cdot\mathfrak{p}_J}_{\mathfrak{p}_I}$$

We note that by proto Langlands formula

$$\mathrm{Res}^{\sigma_1\mathfrak{p}_J\circledcirc\sigma_1\sigma_2\cdot\mathfrak{p}_K}_{\sigma_1\mathfrak{p}_J}\circ\mathrm{Ind}^{\sigma_1\cdot\mathfrak{p}_J}_{\sigma_1\cdot\mathfrak{p}_J\circledcirc\mathfrak{p}_I}=\mathrm{Ind}^{\sigma_1\mathfrak{p}_J\circledcirc\sigma_1\sigma_2\cdot\mathfrak{p}_K}_{\sigma_1\mathfrak{p}_J\circledcirc\sigma_1\sigma_2\cdot\mathfrak{p}_K\circledcirc(\sigma_1\cdot\mathfrak{p}_J\circledcirc\mathfrak{p}_I)}\circ\tau\circ\mathrm{Res}^{\sigma_1\cdot\mathfrak{p}_J\circledcirc\mathfrak{p}_I\circledcirc(\sigma_1\mathfrak{p}_J\circledcirc\sigma_1\sigma_2\cdot\mathfrak{p}_K)}_{\sigma_1\cdot\mathfrak{p}_J\circledcirc\mathfrak{p}_I}$$

We write $\mathfrak{q}=\sigma_1\mathfrak{p}_J\circledcirc\sigma_1\sigma_2\cdot\mathfrak{p}_K$, then

$$\mathrm{Res}^{\sigma_1\cdot\mathfrak{p}_J\circledcirc\mathfrak{p}_I\circledcirc\mathfrak{q}}_{\sigma_1\cdot\mathfrak{p}_J\circledcirc\mathfrak{p}_I}\circ\tau^{\sigma_1\cdot\mathfrak{p}_J\circledcirc\mathfrak{p}_I}_{\mathfrak{p}_I\circledcirc\sigma_1\cdot\mathfrak{p}_J}=\tau^{\sigma_1\cdot\mathfrak{p}_J\circledcirc\mathfrak{p}_I\circledcirc\mathfrak{q}}_{\mathfrak{p}_I\circledcirc\sigma_1\cdot\mathfrak{p}_J\circledcirc\mathfrak{q}}\circ\mathrm{Res}^{\mathfrak{p}_I\circledcirc\sigma_1\cdot\mathfrak{p}_J\circledcirc\mathfrak{q}}_{\mathfrak{p}_I\circledcirc\sigma_1\cdot\mathfrak{p}_J}$$

Note that $\sigma_1\cdot\mathfrak{p}_J\circledcirc\mathfrak{q}=\sigma_1\cdot\mathfrak{p}_J\circledcirc\sigma_1\sigma_2\cdot\mathfrak{p}_K$. The induction side is similar, we have:

$$\tau^{\sigma_1\sigma_2\cdot\mathfrak{p}_K\circledcirc\sigma_1\mathfrak{p}_J}_{\sigma_1\mathfrak{p}_J\circledcirc\sigma_1\sigma_2\cdot\mathfrak{p}_K}\circ\mathrm{Ind}^{\sigma_1\sigma_2\cdot\mathfrak{p}_K\circledcirc\sigma_1\mathfrak{p}_J}_{\sigma_1\sigma_2\cdot\mathfrak{p}_K\circledcirc\sigma_1\mathfrak{p}_J\circledcirc\sigma_1\cdot\mathfrak{p}_J\circledcirc\mathfrak{p}_I}=\mathrm{Ind}^{\sigma_1\sigma_2\cdot\mathfrak{p}_K\circledcirc\sigma_1\mathfrak{p}_J}_{\sigma_1\sigma_2\cdot\mathfrak{p}_K\circledcirc(\sigma_1\cdot\mathfrak{p}_J\circledcirc\mathfrak{p}_I)\circledcirc\sigma_1\mathfrak{p}_J}\,\tau^{\sigma_1\sigma_2\cdot\mathfrak{p}_K\circledcirc(\sigma_1\cdot\mathfrak{p}_J\circledcirc\mathfrak{p}_I)\circledcirc\sigma_1\mathfrak{p}_J}_{\sigma_1\sigma_2\cdot\mathfrak{p}_K\circledcirc\sigma_1\mathfrak{p}_J\circledcirc\sigma_1\cdot\mathfrak{p}_J\circledcirc\mathfrak{p}_I}$$

Taking them back into the right hand side of Proposition, we get

$$\begin{aligned}
&\mathrm{Ind}^{\sigma_1\sigma_2\mathfrak{p}_K}_{\sigma_1\sigma_2\mathfrak{p}_K\circledcirc\sigma_1\cdot\mathfrak{p}_J}\circ\mathrm{Ind}^{\sigma_1\sigma_2\mathfrak{p}_K\circledcirc\sigma_1\cdot\mathfrak{p}_J}_{\sigma_1\sigma_2\mathfrak{p}_K\circledcirc\sigma_1\cdot\mathfrak{p}_J\circledcirc\mathfrak{p}_I}\circ\tau^{\sigma_1\sigma_2\mathfrak{p}_K\circledcirc\sigma_1\cdot\mathfrak{p}_J\circledcirc\mathfrak{p}_I}_{\mathfrak{p}_I\circledcirc\sigma_1\cdot\mathfrak{p}_J\circledcirc\sigma_1\sigma_2\mathfrak{p}_K}\circ\mathrm{Res}^{\mathfrak{p}_I\circledcirc\sigma_1\cdot\mathfrak{p}_J\circledcirc\sigma_1\sigma_2\mathfrak{p}_K}_{\mathfrak{p}_I\circledcirc\sigma_1\sigma_2\mathfrak{p}_K}\,\mathrm{Res}^{\mathfrak{p}_I\circledcirc\sigma_1\sigma_2\mathfrak{p}_K}_{\mathfrak{p}_I}\\
=&\mathrm{Ind}^{\sigma_1\sigma_2\mathfrak{p}_K}_{\sigma_1\sigma_2\mathfrak{p}_K\circledcirc\mathfrak{p}_I}\circ\mathrm{Ind}^{\sigma_1\sigma_2\mathfrak{p}_K\circledcirc\mathfrak{p}_I}_{\sigma_1\sigma_2\mathfrak{p}_K\circledcirc\circledcirc\sigma_1\cdot\mathfrak{p}_J\mathfrak{p}_I}\circ\tau^{\sigma_1\sigma_2\mathfrak{p}_K\circledcirc\circledcirc\sigma_1\cdot\mathfrak{p}_J\mathfrak{p}_I}_{\mathfrak{p}_I\circledcirc\sigma_1\cdot\mathfrak{p}_J\circledcirc\sigma_1\sigma_2\cdot\mathfrak{p}_K}\circ\mathrm{Res}^{\mathfrak{p}_I\circledcirc\sigma_1\cdot\mathfrak{p}_J\circledcirc\sigma_1\sigma_2\cdot\mathfrak{p}_K}_{\mathfrak{p}_I\circledcirc\sigma_1\sigma_2\cdot\mathfrak{p}_K}\,\mathrm{Res}^{\mathfrak{p}_I\circledcirc\sigma_1\sigma_2\cdot\mathfrak{p}_K}_{\mathfrak{p}_I}\\
=&\mathrm{Ind}^{\sigma_1\sigma_2\mathfrak{p}_K}_{\sigma_1\sigma_2\mathfrak{p}_K\circledcirc\mathfrak{p}_I}\circ\tau^{\sigma_1\sigma_2\mathfrak{p}_K\circledcirc\mathfrak{p}_I}_{\mathfrak{p}_I\circledcirc\sigma_1\sigma_2\mathfrak{p}_K}\circ\mathrm{Res}^{\mathfrak{p}_I\circledcirc\sigma_1\sigma_2\mathfrak{p}_K}_{\mathfrak{p}_I}
\end{aligned}$$

it matches the LHS of the Proposition, as we want. ☐

We can see from the construction, when $J\subseteq K$, the double coset represented by the trivial element in $W_J\backslash W/W_K$ gives the induction map $\mathrm{Ind}^{\mathfrak{p}_K}_{\mathfrak{p}_J}$ by taking $\mathfrak{P}_1(\mathfrak{p}_J)$. Similarly, the double coset represented by the trivial element in $W_K\backslash W/W_J$ gives the restriction map. In general, the family of morphism $\tilde{\mathfrak{P}}_\sigma$ can be regarded as a morphism in $\boldsymbol{C}$. As a Corollary of the above Proposition, we have:

Corollary 16. *There is a faithful functor $\boldsymbol{\mathcal{W}}\longrightarrow\boldsymbol{C}$.*

Proof: Their objects are the same, the functor sends $\sigma\in W_J\backslash W/W_K\mapsto\tilde{\mathfrak{P}}_\sigma\in\mathrm{Hom}_{\boldsymbol{C}}(J,K)$. The composition of morphisms in $\boldsymbol{\mathcal{W}}$ sends composition in $\boldsymbol{C}$, due to the above Proposition. ☐

We can regard $\boldsymbol{\mathcal{W}}$ as the categorification of Coxeter group, or the Hecke algebroid specified at $q=1$. Another choice of the specification of q is $q\mapsto 0$. In this situation, the quadratic relation becomes the idempotency on the generators $T_s^2=T_s$. The resulting algebra $\mathcal{H}^{q=0}_{(W,S)}$ is called nil-Hecke algebra. We have the following claim:

Claim: The nil-Hecke algebra can be regarded as a perverse sheaf on $\mathfrak{h}//W$.

The way we show this is to relate the perverse Coxeter category $\boldsymbol{C}$ with a categorification of nil-Hecke algebra, called the *Singular Coxeter category* $\mathcal{SC}_{(W,S)}$, introduced in [EK23]. Similar to $\boldsymbol{\mathcal{W}}$, the objects of $\mathcal{SC}_{(W,S)}$ are $I\subset S$ and morphisms are in bijection with the double coset $\mathrm{Hom}_{\mathcal{SC}}(I,J)=W_J\backslash W/W_I$. Let W_KxW_J and W_JyW_I be two double cosets, their composition in $\mathcal{SC}_{(W,S)}$ is defined to be $W_K\bar{x}*\bar{y}W_I$, where $\bar{x}$ represents the maximal element in the class $W_K x W_J$ and $*$ is the Coxeter monoidal multiplication, satisfying the relations:

$$s * s = s, \quad \forall s \in S;$$
$$\underbrace{s * t * s \ldots}_{m_{s,t}} = \underbrace{t * s * t \ldots}_{m_{s,t}}, \quad \forall s \neq t \in S$$

Remark 17. The relations in Coxeter monoid exhibit the Singular Coxeter category as 0-Hecke category. However, the composition of morphisms in $\mathcal{SC}_{(W,S)}$ is more subtle than the one we have seen in $\mathcal{W}$ due to the fact that $\sigma_1 * \sigma_2 \neq \sigma_1 \cdot \sigma_2$. We are going to see how to interpret morphisms in $\mathcal{SC}$ in our language.

3.2 Type A case and diagrammatic description of $\mathcal{Q}$

We start with the type A case. In [EK23], they establish an equivalence between the collection of singular Coxeter category $\mathcal{SC}(\mathfrak{S}_\star)$ of type A and the category of *Webs*, which has a nice diagrammatic description.
More precisely, the objects in $\mathcal{SC}_{\mathrm{Webs}}$ and $\mathcal{SC}(\mathfrak{S}_\star)$ are ordered partitions $\mathbf{I_n} = (\mathbf{n_1}, \ldots, \mathbf{n_k})$ with $n_i \in \mathbb{N}$ and $\sum_i n_i = n$. We can decompose $\mathcal{SC}_{\mathrm{Webs}}$ as $\bigoplus_n \mathcal{SC}_{\mathrm{Webs}}[n]$, where $\mathrm{Obj}\,(\mathcal{SC}_{\mathrm{Webs}}[n]) = \{\mathbf{I_n}\}$ are those ordered partition of n and in category $\mathcal{SC}_{\mathrm{Webs}}$ we have $\mathrm{Hom}_{\mathcal{SC}_{\mathrm{Webs}}}(\mathbf{I_n}, \mathbf{J_m}) = \emptyset$ if $n \neq m$. The morphisms in $\mathcal{SC}_{\mathrm{Webs}}$ are generated by $\mu^{n+m}_{n,m}$ and $\Delta^{n,m}_{n+m}$ as in Figure 3.

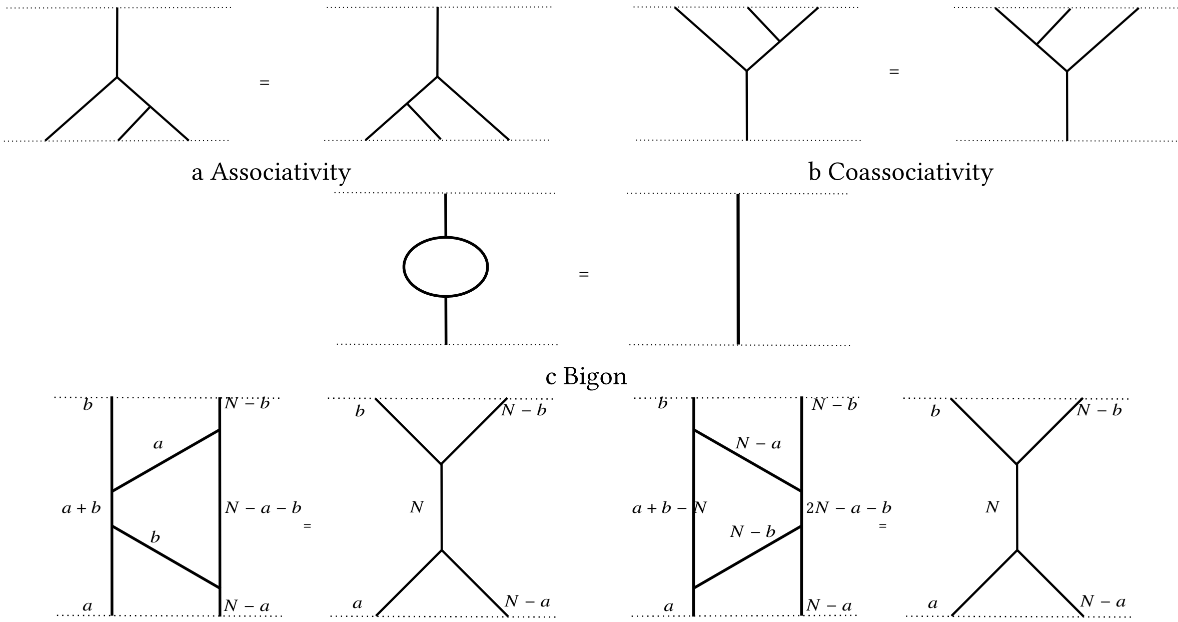
a Induction $\mu^{n+m}_{n,m}$ b Restriction $\Delta^{n,m}_{n+m}$

Figure 3: Ind and Res

modulo the relations as the following Figure 4:
In the type A case, for a fixed Coxeter group $(\mathfrak{S}_n, I_n = \{1, \ldots, n-1\})$, subsets J of I_n are in bijection with ordered partitions $\mathbf{J} = (\mathbf{I_{n_1}}, \ldots, \mathbf{I_{n_k}})$. They coincide with all elements in $\mathcal{L}_{GL(n),T}$ and coincides with elements in $\mathcal{SC}_{\mathrm{Webs}}[n]$. Notice that the condition $J' \subset J \subset I_n$ is equivalent to that the corresponding $\mathbf{J}'$ is a subpartition of $\mathbf{J}$. Also $|J| - |J'|$ is the difference of the partitions $\mathbf{J}$ and $\mathbf{J}'$. We use J to denote the corresponding subset of $\mathbf{J}$.
We now fix $n \in \mathbb{N}$ and compare the webs in degree n with morphisms in the double incidence category $\mathcal{Q}_n = \mathcal{Q}_{GL(n)}$. For any $\mathfrak{p}, \mathfrak{q} \in \mathcal{Q}_n$, if $\mathfrak{p} \preceq \mathfrak{q}$, i.e. $C_\mathfrak{p} \subseteq C_\mathfrak{q}$, then there exists a unique $w \in W = \mathfrak{S}_n$ such that $(\mathfrak{p}, \mathfrak{q}) = w \cdot (\mathfrak{p}_J, \mathfrak{p}_K)$ and $K \subset J$ subsets of $\{1, \ldots, n-1\} = S$. Assume that $|J| - |K| = \ell$, then $\mathfrak{p} \preceq \mathfrak{q}$ can be decomposed as codimension 1 sequence

$$\mathfrak{p} = \mathfrak{p}_0 \preceq_1 \mathfrak{p}_1 \preceq_1 \ldots \preceq_1 \mathfrak{p}_\ell = \mathfrak{q}$$

Here $\mathfrak{p}_i \preceq_1 \mathfrak{p}_{i+1}$ means $\mathrm{codim}_{C_{\mathfrak{p}_{i+1}}}(C^\circ_{\mathfrak{p}_i}) = 1$. Moreover, twist this sequence by w^{-1}, we get $\mathfrak{p}_J = \mathfrak{p}_{J_0} \preceq_1 \mathfrak{p}_{J_1} \preceq_1 \ldots \preceq_1 \mathfrak{p}_{J_\ell} = \mathfrak{p}_K$, where J_{i+1} is obtained by deleting exactly one element in J_i. The codimension 1 sequences between $\mathfrak{p}, \mathfrak{q}$ are in bijection with codimension 1 sequences between the index sets J and K. Moreover, if $J \supset J'$ is codimension 1, then their corresponding ordered partitions are of the form $\mathbf{J} = (\mathbf{I_{n_1}}, \ldots, \mathbf{I_{n_s}}), \mathbf{J}' =$



a Associativity b Coassociativity

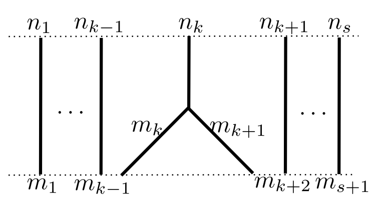
c Bigon

d Switch-back $a+b \leqslant N$ e Switch-back $a+b \geqslant N$

Figure 4: Relations of Webs

$(\mathbf{I_{m_1}}, \dots, \mathbf{I_{m_{s+1}}})$ such that for some $k = 1, \dots, s$, $n_k = m_k + m_{k+1}$, $n_i = m_i$, for $i < k$ and $n_i = m_{i+1}$, for $i > k$. In that case, we consider the diagram:

$$\mu^{J}_{J'} = \mu^{\mathbf{J}}_{\mathbf{J'}} =$$



In the dominant chamber $C_{\mathfrak{p}_\emptyset}$, the above diagram can be regard as the morphism $\mathrm{Ind}^{\mathfrak{p}_J}_{\mathfrak{p}_{J'}}$. For any other facet $\mathfrak{p}$ of type J', there exists a unique facet $C^\circ_{\mathfrak{q}} \subseteq C_{\mathfrak{p}}$ of type J, i.e. $(\mathfrak{p}, \mathfrak{q}) = w \cdot (\mathfrak{p}_{J'}, \mathfrak{p}_J)$ for some w. Over such $\mathfrak{p}$, the above diagram can be regard as the morphism $\mathrm{Ind}^{\mathfrak{q}}_{\mathfrak{p}}$. We thus associate the above diagram with family of morphisms $\mu^{\mathbf{J}}_{\mathbf{J'}} = \left\{ w \cdot \mathrm{Ind}^{\mathfrak{p}_\mathbf{J}}_{\mathfrak{p}_{\mathbf{J'}}} \right\}_{w \in W/W_{J'}}$.
For $\mathfrak{p}_1 \preceq \mathfrak{p}_2 \preceq \mathfrak{p}_3$, there exsits $\mathbf{J_1}, \mathbf{J_2}, \mathbf{J_3}$ such that $\mathbf{J_{i+1}}$ is a subpartition of $\mathbf{J_i}$ and some $w \in W/W_{J_3}$ with $\mathfrak{p}_i = w \cdot \mathfrak{p}_{\mathbf{J_i}}$. Let w' be the image of w in W/W_{J_2}, then $(\mathfrak{p}_1, \mathfrak{p}_2)$ also equals to $w' \cdot (\mathfrak{p}_{\mathbf{J_1}}, \mathfrak{p}_{\mathbf{J_2}})$. We have the equality $\mathrm{Ind}^{\mathfrak{p}_1}_{\mathfrak{p}_3} = \mathrm{Ind}^{\mathfrak{p}_1}_{\mathfrak{p}_2} \circ \mathrm{Ind}^{\mathfrak{p}_1}_{\mathfrak{p}_2}$ in Q. This equality can also be written as

$$\mu^{\mathbf{J_1}}_{\mathbf{J_3}}(\mathfrak{p}_{\mathbf{J_3}}) = w \cdot \mathrm{Ind}^{\mathfrak{p}_{\mathbf{J_1}}}_{\mathfrak{p}_{\mathbf{J_3}}} = (w' \cdot \mathrm{Ind}^{\mathfrak{p}_{\mathbf{J_1}}}_{\mathfrak{p}_{\mathbf{J_2}}}) \circ (w \cdot \mathrm{Ind}^{\mathfrak{p}_{\mathbf{J_2}}}_{\mathfrak{p}_{\mathbf{J_3}}}) = \mu^{\mathbf{J_1}}_{\mathbf{J_2}}(\mu^{\mathbf{J_2}}_{\mathbf{J_3}}(\mathfrak{p}_{\mathbf{J_3}}))$$

The composition of μ's is given in tautological way, by matching the starting facet with the previous target facet. Thus the right hand side of above can be also written as $\mu^{\mathbf{J_1}}_{\mathbf{J_2}} \circ \mu^{\mathbf{J_2}}_{\mathbf{J_3}}(\mathfrak{p}_3)$. It does not depend on the middle partition $\mathbf{J_2}$, hence we get the associativity of μ, as is depicted in Figure 3(a).
Next we consider the morphisms associated with Δ. Assume $\mathbf{J'}$ is a subpartition of $\mathbf{J}$, then for the facet $\mathfrak{p}_{\mathbf{J}}$, there

are $|W_J/W_{J'}|$-facets of type J' adjacent to it. We associate $\Delta_{\mathbf{J}}^{\mathbf{J}'}$ with a particular choice of restriction morphism between facets. Concretely, we set

$$\Delta_{\mathbf{J}}^{\mathbf{J}'} = \left\{ w \cdot \mathrm{Res}_{\mathfrak{p}_{\mathbf{J}}}^{w_{\mathbf{J}}^{\mathbf{J}'} \cdot \mathfrak{p}_{\mathbf{J}'}} \right\}_{w \in W/W_J}$$

where $w_{\mathbf{J}}^{\mathbf{J}'}$ is the maximal element in $W_J/W_{J'}$.

To show coassociativity of the composition of Δ's, we consider $J_1 \supseteq J_2 \supseteq J_3$, namely $\mathbf{J_{i+1}}$ is a subpartition of $\mathbf{J_i}$, then we have $w_{\mathbf{J_1}}^{\mathbf{J_3}} = w_{\mathbf{J_1}}^{\mathbf{J_2}} \cdot w_{\mathbf{J_2}}^{\mathbf{J_3}}$. Apply this to the definition of Δ, we see:

$$\Delta_{\mathbf{J_1}}^{\mathbf{J_3}}(\mathfrak{p}_{\mathbf{J_1}}) = \mathrm{Res}_{\mathfrak{p}_{\mathbf{J_1}}}^{w_{\mathbf{J_1}}^{\mathbf{J_3}} \cdot \mathfrak{p}_{\mathbf{J_3}}} = \mathrm{Res}_{w_{\mathbf{J_1}}^{\mathbf{J_2}} \cdot \mathfrak{p}_{\mathbf{J_2}}}^{w_{\mathbf{J_1}}^{\mathbf{J_2}} \cdot (w_{\mathbf{J_2}}^{\mathbf{J_3}} \cdot \mathfrak{p}_{\mathbf{J_2}})} \circ \mathrm{Res}_{\mathfrak{p}_{\mathbf{J_1}}}^{w_{\mathbf{J_1}}^{\mathbf{J_2}} \cdot \mathfrak{p}_{\mathbf{J_2}}} = \Delta_{\mathbf{J_2}}^{\mathbf{J_3}}(\Delta_{\mathbf{J_1}}^{\mathbf{J_2}}(\mathfrak{p}_{\mathbf{J_1}}))$$

Moreover, the above equalities are compatible with the action of $\sigma \in W/W_{J_1}$, hence we get the coassociativity of Δ, as is depicted in Figure 3 (b).

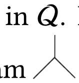
Remark 18. Given a family of morphisms in Q so that different morphihsms have different starting facets, it is the same as a presentation by sum. For example we can write

$$\mu_{\mathbf{J}'}^{\mathbf{J}} = \sum_{w \in W/W_J'} w \cdot \mathrm{Ind}_{\mathfrak{p}_{\mathbf{J}'}}^{\mathfrak{p}_{\mathbf{J}}} \text{ and } \Delta_{\mathbf{J}}^{\mathbf{J}'} = \sum_{w \in W/W_J} w \cdot \mathrm{Res}_{\mathfrak{p}_{\mathbf{J}}}^{w_{\mathbf{J}}^{\mathbf{J}'} \cdot \mathfrak{p}_{\mathbf{J}'}}$$

The composition of μ's and Δ's are thus the composition of the sum. For example when $J_1 \supset J_2 \supset J_3$, let p_{23} be the projection $W/W_{J_3} \to W/W_{J_2}$, then the associativity follows from

$$\begin{aligned}
\mu_{\mathbf{J_2}}^{\mathbf{J_1}} \circ \mu_{\mathbf{J_3}}^{\mathbf{J_2}} &= \left(\sum_{\sigma \in W/W_{J_2}} \sigma \cdot \mathrm{Ind}_{\mathfrak{p}_{\mathbf{J_2}}}^{\mathfrak{p}_{\mathbf{J_1}}} \right) \circ \left(\sum_{\nu \in W/W_{J_3}} \nu \cdot \mathrm{Ind}_{\mathfrak{p}_{\mathbf{J_3}}}^{\mathfrak{p}_{\mathbf{J_2}}} \right) \\
&= \sum_{\sigma \in W/W_{J_2}} \mathrm{Ind}_{\sigma \cdot \mathfrak{p}_{J_2}}^{\sigma \cdot \mathfrak{p}_{J_1}} \circ \left(\sum_{\nu \in p_{23}^{-1}(\sigma)} \mathrm{Ind}_{\nu \cdot \mathfrak{p}_{\mathbf{J_3}}}^{\nu \cdot \mathfrak{p}_{\mathbf{J_2}}} \right) = \sum_{\sigma \in W/W_{J_2}, \nu \in p_{23}^{-1}(\sigma)} \mathrm{Ind}_{\nu \cdot \mathfrak{p}_{J_3}}^{\nu \cdot \mathfrak{p}_{J_1}} \\
&= \sum_{w \in W/W_{J_3}} w \cdot \mathrm{Ind}_{\mathfrak{p}_{J_3}}^{\mathfrak{p}_{J_1}} = \mu_{\mathbf{J_3}}^{\mathbf{J_1}}.
\end{aligned}$$

The coassociativity can also be determined in the similar way.

We have associated the generating diagrams μ and Δ with family of morphisms in Q. To see how other relations can be reflected in Q, we start by associating the Bigon relation (c) in Figure 3 with morphisms in Q. By our previous construction, the diagram  means the morphism $\mathrm{Res}_{\mathfrak{g}}^{w_{[n]}^{\mathbf{J}} \mathfrak{p}_{\mathbf{J}}}$. And the induction diagram  means $\left\{ \mathrm{Ind}_{\sigma \cdot \mathfrak{p}_{\mathbf{J}}}^{\mathfrak{g}} \right\}_{\sigma \in W/W_{\mathbf{J}}}$. Thus, the Bigon relation just says that $\mu_{\mathbf{J}}^{[n]} \circ \Delta_{[n]}^{\mathbf{J}} = \mathrm{Id}_{\mathfrak{g}}$. NB: this is not included in the generating relations of Q.

Let Q_n^{nil} be the quotient category of Q_n modulo furthur relations

$$\mathrm{Ind}_{\mathfrak{p}_{\mathbf{K}}}^{\mathfrak{p}_{\mathbf{J}}} \circ \mathrm{Res}_{\mathfrak{p}_{\mathbf{J}}}^{\mathfrak{p}_{\mathbf{K}}} = \mathrm{id}_{\mathfrak{p}_{\mathbf{J}}}, \quad \text{where } \mathbf{K} \text{ is a subpartition of } \mathbf{J}$$

Theorem 19. *Let $\mathcal{NC}_n$ be the category with objects* $\mathbf{I}$ *partitions of* $[n]$ *and morphisms are generated by* $\mu^{\mathbf{J}}_{\mathbf{K}}$ *and* $\Delta^{\mathbf{K}}_{\mathbf{J}}$ *in* Q^{nil}_n*, where* $\mathbf{K}$ *is a subpartition of* $\mathbf{J}$*. Then it is isomorphic to the category* $\mathcal{SC}_{\mathrm{webs}}[n]$ *and hence to* $\mathcal{SC}_n$*.*

Proof: The asscociativiy and coassociativity are verified. The Bigon relation is obained by definition of Q^{nil}_n. We need to verify the Switch-back relation Figure 3 (d), (e). Let $\mathbf{J}, \mathbf{K}$ correspond to ordered partition $\mathbf{J} = (\mathbf{p}, \mathbf{q}), \mathbf{K} = (\mathbf{r}, \mathbf{s})$, then $W_{\mathbf{J}} \backslash W / W_{\mathbf{K}} = \mathfrak{S}_p \times \mathfrak{S}_q \backslash \mathfrak{S}_n / \mathfrak{S}_r \times \mathfrak{S}_s$ as is shown in [KS25], it is in bijection with the set of contingency matrices

$$\left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \mid a + b = p, c + d = q, a + c = r, b + d = s \right\}$$

For any $\sigma \in W_{\mathbf{J}} \backslash W / W_{\mathbf{K}}$ that corresponds to $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$, we have $\mathbf{J}^{\sigma} = (\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{d})$ and ${}^{\sigma}\mathbf{K} = (\mathbf{a}, \mathbf{c}, \mathbf{b}, \mathbf{d})$. There is a unique longest element σ_0 in $W_{\mathbf{J}} \backslash W / W_{\mathbf{K}}$ (we identify this double coset as the subset of W containing minimal representatives), which corresponds to the contingency matrix of the form

$$\begin{pmatrix} r - q & s \\ q & 0 \end{pmatrix} \quad r \geqslant q, \qquad \begin{pmatrix} 0 & p \\ r & q - r \end{pmatrix} \quad r \leqslant q$$

Note $\mathbf{J}^{\boldsymbol{\sigma_0}} = (\mathbf{r} - \mathbf{q}, \mathbf{s}, \mathbf{q})$ or $(p, r, q - r)$ (depends on whether $r \geqslant q$), it the partition in Figure 3(d),(e). So to show that μ, Δ satisfies the Switch-back relation, we want to show:

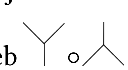
$$\Delta^{\mathbf{K}}_{[n]} \circ \mu^{[n]}_{\mathbf{J}} = \mu^{\mathbf{K}}_{\sigma_0 \mathbf{K}} \circ \Delta^{\sigma_0 \mathbf{K}}_{\mathbf{I}} \circ \mu^{\mathbf{I}}_{\mathbf{J}^{\sigma_0}} \circ \Delta^{\mathbf{J}^{\boldsymbol{\sigma_0}}}_{\mathbf{J}}$$

where $\mathbf{I} = (\mathbf{r} - \mathbf{q}, \mathbf{s} + \mathbf{q})$ or $(p + r, q - r)$, depends on whether $r \geqslant q$. The right hand side is

$$\left(\sum_{\eta \in W / W_{\sigma_0 K}} \eta \cdot \mathrm{Ind}^{\mathfrak{p}_{\mathbf{K}}}_{\mathfrak{p}_{\sigma_0 \mathbf{K}}} \right) \circ \left(\sum_{\gamma \in W / W_I} \gamma \cdot \mathrm{Res}^{w_{\mathbf{I}}^{\sigma_0 \mathbf{K}} \mathfrak{p}_{\sigma_0 \mathbf{K}}}_{\mathfrak{p}_{\mathbf{I}}} \right) \circ \left(\sum_{\sigma \in W / W_{J^{\sigma_0}}} \sigma \cdot \mathrm{Ind}^{\mathfrak{p}_{\mathbf{I}}}_{\mathfrak{p}_{\mathbf{J}^{\boldsymbol{\sigma_0}}}} \right) \circ \left(\sum_{w \in W / W_J} w \cdot \mathrm{Res}^{w_{\mathbf{J}}^{\mathbf{J}^{\boldsymbol{\sigma_0}}} \mathfrak{p}_{\mathbf{J}^{\boldsymbol{\sigma_0}}}}_{\mathfrak{p}_{\mathbf{J}}} \right)$$

We only need to verify that the above map applies to any $w \cdot \mathfrak{p}_{\mathbf{J}}, w \in W / W_J$, we get $w^{\mathbf{K}}_0 \cdot \mathfrak{p}_{\mathbf{K}}$.
We start with $w \mathfrak{p}_{\mathbf{J}}$, then arrive at $w w_{\mathbf{J}}^{\mathbf{J}^{\boldsymbol{\sigma_0}}} \mathfrak{p}_{\mathbf{J}^{\boldsymbol{\sigma_0}}}$. Since $W / W_J \subseteq W / W_{J^{\sigma_0}}$, there is a unique $\sigma \in W / W_{J^{\sigma_0}}$ so that the first composition is non-trivial. Then we can conclude with the following lemma:

Lemma 20. *For the partitions above, we have:*
(i) $w_{\mathbf{J}}^{\mathbf{J}^{\boldsymbol{\sigma_0}}} \cdot w_{\mathbf{I}}^{\sigma_0 \mathbf{K}} = w^{\mathbf{K}}_0 = w^{\mathbf{K}}_{[n]}$.
(ii) The image of $(W / W_J) \cdot w_{\mathbf{J}}^{\mathbf{J}^{\boldsymbol{\sigma_0}}}$ *under the projection map* $W / W_{J^{\sigma_0}} \longrightarrow W / W_I$ *lies in the set* $W_{\sigma_0 K} \cdot (w_{\mathbf{J}}^{\mathbf{J}^{\boldsymbol{\sigma_0}}})$.

Example 21. Let us first consider the simplest case A_1, where the corresponding group is GL_2. The web  in this case is thus $\Delta^{(1,1)}_{[2]} \circ \mu^{[2]}_{(1,1)}$. The formula for it in Q^{nil}_2 is

$$\mathrm{Res}^{s_\alpha \cdot \mathfrak{b}}_{\mathfrak{g}} \circ \left(\mathrm{Ind}^{\mathfrak{g}}_{\mathfrak{b}} + \mathrm{Ind}^{\mathfrak{g}}_{s_\alpha \cdot \mathfrak{b}} \right) = \mathrm{Res}^{s_\alpha \cdot \mathfrak{b}}_{\mathfrak{g}} \circ \mathrm{Ind}^{\mathfrak{g}}_{\mathfrak{b}} + \mathrm{Id}_{s_\alpha \cdot \mathfrak{b}} .$$

Note also by Bigon relation, we have the following idempotency, Figure 5:
This idempotency can also be obtained from composing $\mathrm{Res}^{s_\alpha \cdot \mathfrak{b}}_{\mathfrak{g}} \circ \mathrm{Ind}^{\mathfrak{g}}_{\mathfrak{b}} + \mathrm{Id}_{s_\alpha \cdot \mathfrak{b}}$ with itself.

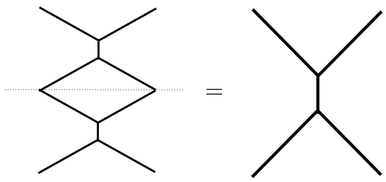


Figure 5: The idempotent morphism $(\Delta^{(1,1)}_{[2]} \circ \mu^{[2]}_{(1,1)}) \circ (\Delta^{(1,1)}_{[2]} \circ \mu^{[2]}_{(1,1)}) = \Delta^{(1,1)}_{[2]} \circ \mu^{[2]}_{(1,1)}$

Example 22. Let us then consider the case of GL_3 (type A_2). Let α, β be the two vertices of the Dynkin diagram. The Coxeter complex is demonstrated as in Figure 1. The picture 6 exhibits some morphisms in $\mathcal{Q}^{\mathrm{nil}}_3$.

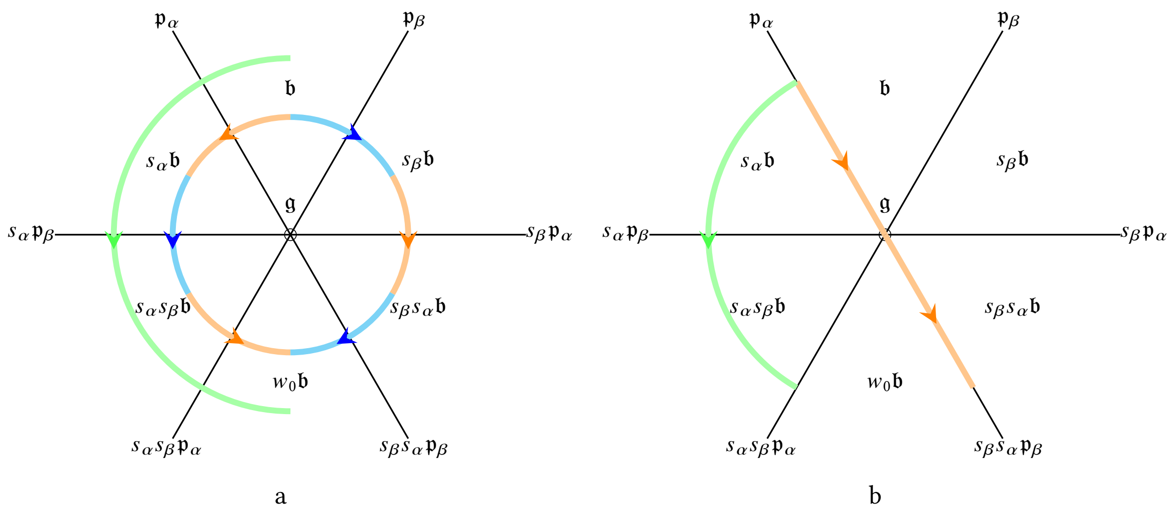
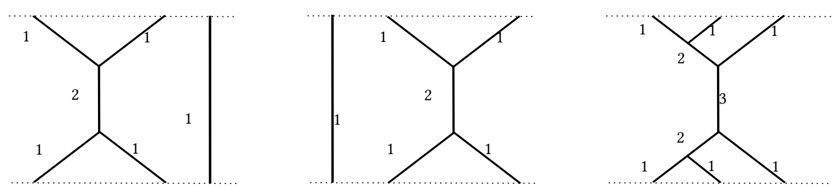


Figure 6: Type A_2

Let us assume α corresponds to the partition $(2,1)$ and β corresponds to the partition $(1,2)$. In Figure 6(a), the orange path $\mathfrak{b} \longrightarrow s_\alpha\mathfrak{b}$ is the map $\mathrm{Res}^{s_\alpha\mathfrak{b}}_{\mathfrak{p}_\alpha} \circ \mathrm{Ind}^{\mathfrak{p}_\alpha}_{\mathfrak{b}}$. It can also be regarded as $\Delta^{(1,1,1)}_{(2,1)} \circ \mu^{(2,1)}_{(1,1,1)}(\mathfrak{b})$. The blue path $\mathfrak{b} \longrightarrow s_\beta\mathfrak{b}$ is the map $\mathrm{Res}^{s_\beta\mathfrak{b}}_{\mathfrak{p}_\beta} \circ \mathrm{Ind}^{\mathfrak{p}_\beta}_{\mathfrak{b}} = \Delta^{(1,1,1)}_{(1,2)} \circ \mu^{(1,2)}_{(1,1,1)}(\mathfrak{b})$. And finally according to the proto Langlands formula, the green path $\mathfrak{b} \longrightarrow w_0\mathfrak{b}$ represents the map $\mathrm{Res}^{w_0\mathfrak{b}}_{\mathfrak{g}} \circ \mathrm{Ind}^{\mathfrak{g}}_{\mathfrak{b}} = \Delta^{(1,1,1)}_{[3]} \circ \mu^{[3]}_{(1,1,1)}(\mathfrak{b})$.



a $\Delta^{(1,1,1)}_{(2,1)} \circ \mu^{(2,1)}_{(1,1,1)}$ b $\Delta^{(1,1,1)}_{(1,2)} \circ \mu^{(1,2)}_{(1,1,1)}$ c $\Delta^{(1,1,1)}_{[3]} \circ \mu^{[3]}_{(1,1,1)}$

Figure 7: diagrams in A_2 represent $\emptyset \longrightarrow \emptyset$

As in Figure 8, As is depicted in the Figure 6(a)we can deduce that the green path is the composition of orange-blue-orange or blue-orange-blue paths. This can also be obtained by applying switch-back relation to Figure 7(c)(and valuing at $\mathfrak{b}$).

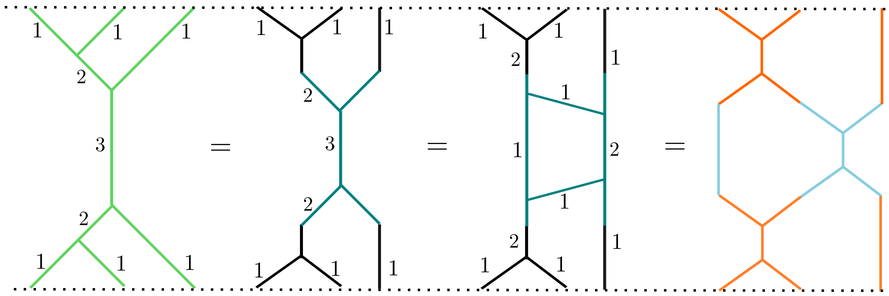


Figure 8: diagrammatic description of equivalence of paths in A_2

The diagrammatic computation is shown in Figure 8. In fact, we can also compute directly

$$\begin{aligned}
&\left(\Delta^{(1,1,1)}_{(2,1)} \circ \mu^{(2,1)}_{(1,1,1)}\right) \circ \left(\Delta^{(1,1,1)}_{(1,2)} \circ \mu^{(1,2)}_{(1,1,1)}\right) \circ \left(\Delta^{(1,1,1)}_{(2,1)} \circ \mu^{(2,1)}_{(1,1,1)}\right) \\
=& \left(\sum_{w \in \mathfrak{S}_3/W_\alpha} \mathrm{Res}^{ws_\alpha \cdot \mathfrak{b}}_{w \cdot \mathfrak{p}_\alpha}\right) \circ \left(\sum_{v \in \mathfrak{S}_3} \mathrm{Ind}^{v \cdot \mathfrak{p}_\alpha}_{v \cdot \mathfrak{b}}\right) \circ \left(\sum_{w' \in \mathfrak{S}_3/W_\beta} \mathrm{Res}^{w's_\beta \cdot \mathfrak{b}}_{w' \cdot \mathfrak{p}_\beta}\right) \circ \left(\sum_{v' \in \mathfrak{S}_3} \mathrm{Ind}^{v' \cdot \mathfrak{p}_\beta}_{v' \cdot \mathfrak{b}}\right) \\
&\circ \left(\sum_{w \in \mathfrak{S}_3/W_\alpha} \mathrm{Res}^{ws_\alpha \cdot \mathfrak{b}}_{w \cdot \mathfrak{p}_\alpha}\right) \circ \left(\sum_{v \in \mathfrak{S}_3} \mathrm{Ind}^{v \cdot \mathfrak{p}_\alpha}_{v \cdot \mathfrak{b}}\right) = \sum_{\sigma \in \mathfrak{S}_3} \mathrm{Res}^{w_0 \mathfrak{b}}_{\mathfrak{g}} \circ \mathrm{Ind}^{\mathfrak{g}}_{\sigma \cdot \mathfrak{b}} = \Delta^{(1,1,1)}_{[3]} \circ \mu^{[3]}_{(1,1,1)}
\end{aligned}$$

Also we can see the orange path in Figure 6(b) is the morphism $\Delta^{(1,2)}_{[3]} \circ \mu^{[3]}_{(2,1)}(\mathfrak{p}_\alpha)$ and the green path in Figure 6(b) is the morphism $\Delta^{(2,1)}_{[3]} \circ \mu^{[3]}_{(2,1)}(\mathfrak{p}_\alpha)$.

3.3 The Coxeter monoid for reductive groups

Now we move to the general case. Let (W, S) be any finitary Coxeter system. We have mentioned there are two monoidal structures $*$ and $\cdot$ on W, which can be regarded as Iwahori-Hecke algebra specialized at 0 and 1 respectively. By definition, we can see for any word in W, it is a reduced one under $\cdot$ if and only if it is reduced under $*$. To lift the Coxeter monoid structure to the catenation of double cosets, in [EK23], the authors study the category $\mathcal{SC}$ through the presentation of double cosets in (reduced) singular sequences and determine their relations. We recap their construction: by *singular multisetp sequence* we mean a sequence of subsets of S of the form

$$L_\bullet = [I_0 \subset K_1 \supset I_1 \subset \ldots \subset K_m \supset I_m]$$

The inclusions can be non-strict. One can associate $L_\bullet$ with an element $w_{L_\bullet}$ in W:

$$w_{L_\bullet} := w_{K_1} w_{I_1}^{-1} \ldots w_{I_{m-1}}^{-1} w_{K_m}, \text{ where } w_K \in W \text{ is the maximal length representative in } W/W_K$$

We say a singular sequence $L_\bullet$ as above is *reduced* if $\ell(w_{L_\bullet}) = \ell(w_{K_1}) - \ell(w_{I_1}) + \ldots + \ell(w_{K_m})$, where $\ell(\bullet)$ is the usual length function for Coxeter group $(W, \cdot)$. In particular let $p = W_{I_0} w_{L_\bullet} W_{I_m}$ be the corresponding (I_0, I_m)-coset. If $L_\bullet$ is reduced then we have $\bar{p}$ the representative of maximal length in p, equals to $w_{L_\bullet}$, in such case, we write $L_\bullet \rightleftharpoons p$. If two reduced sequence $L_\bullet, L'_\bullet$ both start with I_0, end with I_m and $L_\bullet \rightleftharpoons p \rightleftharpoons L'_\bullet$, we write $L_\bullet \rightleftharpoons L'_\bullet$.

Example 23. Let $J \subset S$ be a proper subset and suppose $s \notin J, t \in J$, then $[J \subset J \cup \{s\}]$ and $[J \supset J \backslash \{t\}]$ are reduced sequences and the double coset contains $w_{J\cup\{s\}}$ (resp. w_J) is in the same class as double coset contains 1. We abbreviate them as $[J + s]$ and $[J - t]$ respectively. In general we write $[J + s_1 - s_2 \ldots]$ for sequences in which each step differs by one element, and we call them *(single step) expressions*. Beware that single-step expressions of steps $\geqslant 2$ are no longer reduced in general.

Lemma 24. *For any double coset $p \in W_J\backslash W/W_I$, there exists a reduced sequence $L_\bullet$ starts with J and ends with I such that $W_J w_{L_\bullet} W_I$ is equivalent to p.*

The above Lemma is due to Williamson in his study of singular Soergel bimodules. Given V a reflection faithful representation of W, let $R = \mathrm{Sym}(V)$ and R^I denote the W_I invariant in R for $I \subset S$, a sequence $L_\bullet = [I_0 \subset K_1 \supset I_1 \subset \ldots \subset K_m \supset I_m]$ gives an (R^{I_0}, R^{I_m})-bimodule

$$R^{I_m} \otimes_{R^{K_m}} R^{I_{m-1}} \otimes \ldots \otimes_{R^{K_1}} R^{I_0}$$

In fact, Williamson constructed the following category of singular Bott-Samelson bimodules $\mathcal{S}\mathbb{BS}_{(W,S)}$ with objects to be subsets $I \subset S$, which are identified with R^I and morphisms are generated by

$$\mathrm{Res}^J_I := {}_{R^J}R^J{}_{R_I}, \quad \mathrm{Ind}^I_J := {}_{R^I}R^J{}_{R^J}, \quad \text{where } J \subset I, |J| = |I| - 1.$$

The composition of morphisms is given by the tensor product. It is clear by definition that $\mathrm{Res}^K_J \circ \mathrm{Res}^J_I = \mathrm{Res}^K_I$ and $\mathrm{Ind}^I_J \circ \mathrm{Ind}^J_K = \mathrm{Ind}^I_K$. So for any singular sequence $L_\bullet = [I_0 \subset \ldots \supset I_m]$, its associated Bott-Samelson (R^{I_0}, R^{I_m})-bimodule is as above.

Example 25. For $(W = \mathfrak{S}_2, S = \{s\})$, the two indecomposable (R, R)-Soergel bimodules are R itself and $B_s = R \otimes_{R^s} R$, the later corresponds to a reduced sequences $[\emptyset, s, \emptyset]$.

Theorem 26 (Williamson)**.** *There is a bijection between indecomposable singular Soergel bimodules in* $\mathrm{Hom}_{\mathcal{S}\mathbb{BS}}(J, K)$ *and the double coset $W_J\backslash W/W_K$. Moreover, the Karoubi completion $\mathcal{S}\mathbb{S} :=$ Kar $\mathcal{S}\mathbb{BS}$ of singular Bott-Samelson bimodules gives a categorification of Hecke algebroid.*

Two singular sequences $L_\bullet = [I_0 \subset K_1 \supset I_1 \subset \ldots \subset K_m \supset I_m]$ and $L'_\bullet = [I'_0 \subset K'_1 \supset I'_1 \subset \ldots \subset K'_{m'} \supset I'_{m'}]$ are composable (by right) if $I_m = I'_0$. If such case, we write their composition to be

$$L_\bullet \circ L'_\bullet := [I_0 \subset K_1 \supset I_1 \subset \ldots \subset K_m \supset I_m = I'_0 \subset K'_1 \ldots \subset K'_{m'} \supset I'_{m'}]$$

Now consider $s \notin J \subseteq S$ and $t \in J$. We write in short Js for $J \cup \{s\}$ and $Js\backslash t$ for $(J \cup \{s\})\backslash \{t\}$. Let $u_0 = s$ and $u_{-1} = w_{Js} t w_{Js}$ be initial elements of the series $\{u_n\}_{n\geqslant -1}$, where $u_{i+1} := w_{I_i} u_{i-1} w_{I_i}, I_i := Js\backslash u_i$. Accordingly, let $L_i = Js\backslash \{u_i, u_{i-1}\}$, we get a singular sequence: $[I_0 = J \supset L_1 \subset I_1 \supset L_2 \ldots I_k \supset L_{k+1} \subset I_{k+1} \ldots]$.

Lemma 27. *[EK23, Lemma 5.11] The sequence $\{u_n\}_{n\geqslant -1}$ is well-defined and if (W, S) is finitary, this sequence is periodic. Moreover, let δ be the largest integer so that $[I_0 \supset L_1 \ldots L_\delta \subset I_\delta]$ is reduced, then $u_\delta = t$ and $u_{\delta+1} = w_{Js} s w_{Js}$.*

Theorem 28. *[EK23, Proposition 5.14, Theorem 5.32] The singular sequences give a presentation of singular Coxeter category with composition as above, under the following relations:*

(1) $[J+s+t] \rightleftharpoons [J+t+s]; [J-s-t] \rightleftharpoons [J-t-s]$.

(2) $[J-s+s] = J$.

(3) (Switch back relations) Given $s \notin J$ *and* $t \in J$*, if*

(i) $s = w_{Js} t w_{Js}$*, then* $[J+s-t]$ *is the unique reduced sequence that* $\leftrightharpoons p$*, where* p *is the coset of* w_{Js} *in* $W_J \backslash W_{Js} / W_{Js\backslash t}$.

(ii) Otherwise $[J+s-t] \leftrightharpoons [J-u_1+s-u_2 \ldots -u_{\delta-1}+u_{\delta-2}-t+u_{\delta-1}]$*, where* δ *is the same integer as in Lemma 27.*

We now relate the construction of Singular Coxeter category with the Coxeter category $\boldsymbol{C}$. In the sequel, for $J \subseteq I$, we always write W_I/W_J the set of minimal representative in W_I. Also we denote by w_I^J the maximal (length) element in W_I/W_J.

For any sequence $L_\bullet = [I_0 \subset K_1 \supset I_1 \subset \ldots \subset K_m \supset I_m]$, we first decompose it into blocks, where each block is of the form $[I_{i-1} \subset K_i \supset I_i]$. It is clear by definition that each block $L^i_\bullet = [I_{i-1} \subset K_i \supset I_i]$ is a reduced singular sequence, let us call it *fundamental singular sequence.*

For such block, we can associate it with a family of morphisms in $\boldsymbol{Q}$ that starts with facet of type I_{i-1}, ends with type I_i and passes type K_i. In analogy to the type A-case, we associate it a family of morphisms in $\boldsymbol{Q}$:

$$\mathfrak{P}_{L^i_\bullet} = \left(\sum_{w \in W/W_{K_i}} w \cdot \mathrm{Res}_{\mathfrak{p}_{K_i}}^{\mathfrak{p}_{K_i} \circledcirc w_{K_i}^{I_{i-1}} \cdot \mathfrak{p}_{I_i}} \right) \circ \left(\sum_{v \in W/W_{I_i}} v \cdot \mathrm{Ind}_{\mathfrak{p}_{I_{i-1}}}^{\mathfrak{p}_{K_i}} \right)$$

In particualr when $L^i_\bullet$ is of the form $[J \subset I = I]$, then $\mathfrak{P}_{L^i_\bullet}$ is the family of induction maps $\left\{ w \cdot \mathrm{Ind}_{\mathfrak{p}_J}^{\mathfrak{p}_I} \right\}_{w \in W/W_J}$. Notice when $J \subset I$, starts with any facet of type J, there is only one facet of type I that ajacent to it, thus there is no ambiguity when choosing the induction path.

When $L^i_\bullet$ is of the form $[I = I \supset J]$, then $\mathfrak{P}_{L^i_\bullet}$ is the family of restriction maps $\left\{ w \cdot \mathrm{Res}_{\mathfrak{p}_I}^{w_I^J \cdot \mathfrak{p}_J} \right\}_{w \in W/W_I}$. Start with the standard facet $\mathfrak{p}_I$, there are $|W_I/W_J|$-facets of type $J \subset I$ that ajacent to $\mathfrak{p}_I$, we here fix the direction of restriction by associating $[I \supset J]$ with one certain path $\mathrm{Res}_{\mathfrak{p}_I}^{w_I^J \cdot \mathfrak{p}_J}$.

Now for $L^1_\bullet = [I_0 \subset K_1 \supset I_1]$ and $L^2_\bullet = [I_1 \subset K_2 \supset I_2]$ two basics blocks that are composable, we can connect two families of morphisms by matching the starting facet of $\mathfrak{P}_{L^2_\bullet}$ with the ending facet of $\mathfrak{P}_{L^1_\bullet}$. The corresponding family of morphisms is associated with $L^1_\bullet \circ L^2_\bullet$, i.e.

$$\mathfrak{P}_{L^1_\bullet \circ L^2_\bullet} = \mathfrak{P}_{L^2_\bullet} \circ \mathfrak{P}_{L^1_\bullet}$$

Under the above composition law, we can associate every singular sequence $L_\bullet$ with a family of morphisms $\mathfrak{P}_{L_\bullet}$ in $\boldsymbol{Q}$. Theorem 28 gives all the equivalence relations of singular sequences, if $L_\bullet \leftrightharpoons L'_\bullet$, the question is whether their associating paths $\mathfrak{P}_{L_\bullet}$ and $\mathfrak{P}_{L'_\bullet}$ will be equivalent families of morphisms in $\boldsymbol{Q}$. It is enough to verify for generating relations of singular sequences.

Let J be any subset of S that does not contain $s \neq t \in S$, then it is easy to verify that

$$\mathfrak{P}_{[J+s+t]} = \sum_{v \in W/W_J} v \cdot (\mathrm{Ind}_{\mathfrak{p}_{J\cup\{s\}}}^{\mathfrak{p}_{J\cup\{s,t\}}} \circ \mathrm{Ind}_{\mathfrak{p}_J}^{\mathfrak{p}_{J\cup\{s\}}}) = \sum_{v \in W/W_J} v \cdot (\mathrm{Ind}_{\mathfrak{p}_J}^{\mathfrak{p}_{[J+s+t]}}) = \mathfrak{P}_{[J+t+s]}$$

To show the compatibility of restrictions, we need the following Lemma

Lemma 29. *Let $K \subseteq J \subseteq I$ be subsets of S, where (W, S) is a given finitary Coxeter system. Then the maximal element w_I^K in W_I/W_K equals $w_I^J \cdot w_J^K$.*

For $I \supset J$, we can write $\mathfrak{P}_{[I\supset J]} = \sum_{w \in W_I} w \cdot \mathrm{Res}_{\mathfrak{p}_I}^{w_I^J \cdot \mathfrak{p}_J}$. For $I \supset J \supset K$, we have

$$\begin{aligned}\mathfrak{P}_{[I\supset J\supset K]} &= \left(\sum_{w \in W/W_J} w \cdot \mathrm{Res}_{\mathfrak{p}_J}^{w_J^K \cdot \mathfrak{p}_K}\right) \circ \left(\sum_{v \in W/W_I} v \cdot \mathrm{Res}_{\mathfrak{p}_I}^{w_I^J \cdot \mathfrak{p}_J}\right) \\ &= \sum_{v \in W/W_I} \sum_{\left\{w \in W/W_J \mid w \cdot \mathfrak{p}_I = v w_I^J \cdot \mathfrak{p}_J\right\}} \mathrm{Res}_{v \cdot \mathfrak{p}_J}^{w w_J^K \cdot \mathfrak{p}_K}\end{aligned}$$

Since we take W/W_I to be the minimal representative in W, thus W/W_I is a subset W/W_J, also $w_I^J \in W_I/W_J \subseteq W/W_J$, thus the index of the second summand is just a singleton $\left\{v w_I^J\right\}$. From above Lemma, we deduce that the above formula is just $\sum_{v \in W/W_I} \mathrm{Res}_{v \cdot \mathfrak{p}_I}^{v w_I^K \cdot \mathfrak{p}_K}$, and thus $\mathfrak{P}_{[I\supset J\supset K]} = \mathfrak{P}_{[I\supset K]}$. In particular, apply this to $[J-s-t]$ and $[J-t-s]$, we can deduce $\mathfrak{P}_{[J-s-t]} = \mathfrak{P}_{[J-t-s]} = \mathfrak{P}_{[J\supset J\backslash\{s,t\}]}$.
Next we consider the case $[J - s + s]$, the associated $\mathfrak{P}_{[J-s+s]} = \mathfrak{P}_{[J\supset J\backslash\{s\}]\circ[J\backslash\{s\}\subseteq J]}$ applies to $\mathfrak{p}_J$ as $\mathrm{Ind}_{w_J^{J\backslash s} \cdot \mathfrak{p}_{J\backslash s}}^{\mathfrak{p}_J} \circ \mathrm{Res}_{\mathfrak{p}_J}^{w_J^{J\backslash s} \cdot \mathfrak{p}_{J\backslash s}}$. It is not the identity map in Q. Thus, similar to the cases of $\mathcal{SC}_{\mathrm{Webs}}$, we need to consider the quotient category Q^{nil} of Q obtained by modulo the relation:

$$\mathrm{Ind}_{w_J \cdot \mathfrak{p}_{J\backslash\{s\}}}^{\mathfrak{p}_J} \circ \mathrm{Res}_{\mathfrak{p}_J}^{w_J \cdot \mathfrak{p}_{J\backslash\{s\}}} = \mathrm{id}_{\mathfrak{p}_J}, \quad \forall s \in J \subseteq S \quad \text{(Bigon relation)}$$

In general, for any $K \subset J \subset S$, write $J\backslash K = \{s_1, \ldots, s_l\}$. Combining the above Bigon relation and the compatibilities of induction/restriction maps above, we have $\mathfrak{P}_{[J\supset K\subset J]} = \mathfrak{P}_{[J-s_1-\ldots-s_l+s_l+\ldots+s_1]}$ is the identity.
Finally, let us consider the Switch-back relations. According to the previous construction, the family of morphisms $\mathfrak{P}_{[J+s-t]}$ is given by

$$\mathfrak{P}_{[J+s-t]} = \left(\sum_{w \in W/W_{Js}} w \cdot \mathrm{Res}_{\mathfrak{p}_{Js}}^{w_{Js}^{Js\backslash t} \cdot \mathfrak{p}_{Js\backslash t}}\right) \circ \left(\sum_{v \in W/W_J} v \cdot \mathrm{Ind}_{\mathfrak{p}_J}^{\mathfrak{p}_{Js}}\right)$$

Note that $\mathfrak{p}_J \circledcirc w_{Js}^{Js\backslash t} \cdot \mathfrak{p}_{Js\backslash t} = \mathfrak{p}_J$ if and only if $s = w_{Js} t w_{Js}$, in which case $C^\circ_{\mathfrak{p}_J}$ and $C^\circ_{w_{Js} \cdot \mathfrak{p}_{Js\backslash t}}$ are in the same flat. So we can see that $\mathrm{Ind}_{w_{Js} \cdot \mathfrak{p}_{Js\backslash t} \circledcirc \mathfrak{p}_J}^{w_{Js} \cdot \mathfrak{p}_{Js\backslash t}}$ and $\mathrm{Res}_{\mathfrak{p}_J}^{\mathfrak{p}_J \circledcirc w_{Js} \cdot \mathfrak{p}_{Js\backslash t}}$ are trivial, and the generic line segment between $C^\circ_{\mathfrak{p}_J}$ and $C^\circ_{w_{Js} \cdot \mathfrak{p}_{Js\backslash t}}$ intersects transversally with $C_{\mathfrak{p}_{Js}}$.
If not in the above case, we want to have $\mathfrak{P}_{[J+s-t]} = \mathfrak{P}_{[J-u_1+s\ldots-t+u_\delta]}$, which is more complicated in general than type A-case. We arrive at the following theorem:

Theorem 30. *Two singular sequences $L_\bullet$ and $L'_\bullet$ are equivalent if and only if their associated $\mathfrak{P}_{L_\bullet}$ and $\mathfrak{P}_{L'_\bullet}$ have every term matches in Q^{nil}. In other words, if we consider the category $\mathcal{NC}$ with objects $I \subseteq S$ and morphisms are generated by $\mathfrak{P}_{[J\subset I]}$ and $\mathfrak{P}_{[I\supset J]}$, then $\mathcal{NC}$ is isomorphic to the singular Coxeter category $\mathcal{SC}$.*

Proof: To show this, we first to show that if two singular sequences $L_\bullet \leftrightharpoons L'_\bullet$ with the same source I and target J, then $\mathfrak{P}_{L_\bullet}(w \cdot \mathfrak{p}_I) = \mathfrak{P}_{L'_\bullet}(w \cdot \mathfrak{p}_I)$ for all $w \in W/W_I$. According to Theorem 28, all equivalence relations are reduced to three types. We only need to show that $\mathfrak{P}_{[J+s-t]} = \mathfrak{P}_{[J-u_1+s\ldots-t+u_\delta]}$. Moreover, one can verify that if $L_\bullet = [I \subseteq K \supseteq J]$ is a fundamental sequence, then $\mathfrak{P}_{L_\bullet}(w \cdot \mathfrak{p}) = w \cdot \mathfrak{P}_{L_\bullet}(\mathfrak{p}), \forall \mathfrak{p}$ with color I. Thus we have

$$\mathfrak{P}_{L^1_\bullet \circ L^2_\bullet}(w \cdot \mathfrak{p}) = \mathfrak{P}_{L^2_\bullet}\left(w \cdot \mathfrak{P}_{L^1_\bullet}(\mathfrak{p})\right) = w \cdot \mathfrak{P}_{L^1_\bullet \circ L^2_\bullet}(\mathfrak{p})$$

And we can deduce for any singular sequence $L_\bullet$, $\mathfrak{P}_{L_\bullet}$ commutes with action of W. Thus, it reduce to test the formula $\mathfrak{P}_{[J+s-t]} = \mathfrak{P}_{[J-u_1+s\ldots-t+u_\delta]}$ at the standard parabolic $\mathfrak{p}_J$, by definition, the left hand side applies to $\mathfrak{p}_J$ gives $w_{Js}^{Js\backslash t} \cdot \mathfrak{p}_{Js\backslash t} = w_{Js} \cdot \mathfrak{p}_{Js\backslash t}$ while the right hand side gives

$$w_J^{J\backslash u_1} w_{Js\backslash u_1}^{Js\backslash u_1 u_2} \ldots w_{Js\backslash u_{\delta-1}}^{Js\backslash u_{\delta-1} u_\delta} \cdot \mathfrak{p}_{Js\backslash u_\delta} = w_J^{J\backslash u_1} w_{Js\backslash u_1}^{Js\backslash u_1 u_2} \ldots w_{Js\backslash u_{\delta-1}}^{Js\backslash u_{\delta-1} u_\delta} \cdot \mathfrak{p}_{Js\backslash t}$$

It remains to show

$$w_{Js} \equiv w_J w_{Js\backslash u_1} \ldots w_{Js\backslash u_\delta} \operatorname{mod} W_{Js\backslash t}$$

which follows from the construction of the sequence $[u_{-1}u_0\ldots u_{\delta-1}u_\delta]$. We then need to show that the orbit category Q^{nil}/W is generated by $\underline{\mathrm{Ind}}_I^J$ and $\underline{\mathrm{Res}}_J^I$ for all $I \subseteq J \subseteq S$. $\square$

In conclusion, we obtain a perverse sheaf on $\mathfrak{h}//W$ that corresponds to nil-Hecke algebra by constructing a functor $\boldsymbol{C} \longrightarrow \mathrm{Vect}$ via the following diagram:

$$\begin{tikzcd}
Q \arrow[rr, two heads] \arrow[d, squiggly] & & Q^{\mathrm{nil}} \arrow[d, squiggly] \\
\boldsymbol{C} = Q/W \arrow[rr, two heads] \arrow[dr] & & \mathcal{SC} = Q^{\mathrm{nil}}/W \arrow[dl, "\text{nil-Hecke}"] \\
& \mathrm{Vect} &
\end{tikzcd}$$

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